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Communicational Problems in Aviation Operation

A practical approach

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Management summary

Large disasters such as Chernobyl, Bhopal and Tenerife are linked to communicational problems. Studies have found that in 70% to 80% of the aviation incidents, communicational problems were a contributing factor. These findings suggest that communication plays a large role in the safety of aviation operation. Studies have presented the frequency of communicational problems that contributed to accidents, but they do not refer to types of communication and kinds of interactions.

The assignment from the Research Department of the Aviation Academy was to make a first attempt to produce an analysis tool for communicational problems. This tool would identify contributing factors, regarding communication, by their kinds of interactions instead of their types of problems. The following research question is answered in this research:

What analysis tool can be used to analyse aviation accident and incident investigation reports, so to identify the kind of interactions that are most frequently stated as contributing factor in these reports, and thereby give insight in what types of communicational problems aviation operation experiences?

The aspects of communication - that were found to analyse the contributing factors - are the actors, distance, timing, flow of information, form, senses involved and communication media. All these aspects were consequently specified for each contributing factor found in 31 aviation accident and incident investigation reports by the Dutch Safety Board. The analysis of the collection of these 103 contributing factors would reveal which kinds of interactions are most frequently present in the investigation reports.

Interactions between ATC controllers and other human actors in aviation operation were often found as contributing factors. Human-document interactions and human-machine interactions are found frequently, indicating that humans in aviation operation have issues understanding the tools they use.

In future research at the Research Department of the Aviation Academy, regarding the production of an analysis tool for communicational problems, a few recommendations are to be considered. In this research was found that an abundant amount of information could be created from the results. It is recommended that the amount of aspects are reduced, as this will further ease the analysis process, improve clarity of analysis tool and deliver more manageable results. A second recommendation that is made, is that the actors in the interaction are specified more accurately by using abbreviations of the kind of actor, instead of only the specification if it is human or not. This will deliver clearer results and would result in a more accurate post analysis.

Management samevatting

Grote rampen zoals Chernobyl, Bhopal en Tenerife zijn gekoppeld aan communicatieproblemen. Onderzoek heeft uitgewezen dat in 70% tot 80% van de luchtvaart incidenten, communicatieproblemen een 'contributing factor' waren. Deze vindingen suggereren dat communicatie een grote rol speelt in de veiligheid van luchtvaart operatie. Voorgaande onderzoeken presenteerden de frequentie van types communicatieproblemen die vaak hebben bijgedragen aan luchtvaart ongelukken en incidenten. Deze onderzoeken refereerden niet naar types communicatie en interacties die vaak worden beschreven als 'contributing factor'.

De Research Department van de Aviation Academy gaf de opdracht om een eerste aanzet te doen om een 'tool' te produceren waarmee communicatieproblemen kunnen worden geanalyseerd. Deze tool zou worden gebruikt om contributing factors, met betrekking tot communicatie, te analyseren en beschrijven op de interactie die ze beschrijven, in plaats van het type communicatieprobleem. De onderzoeksvraag luidt:

Wat voor analyse instrument kan worden gebruikt om luchtvaart ongeluk en incident onderzoeksrapporten te analyseren, om zo de types interacties te identificeren die het meest frequent worden beschreven als contributing factor, en zo inzicht te geven in wat voor communicatieproblemen luchtvaart ondervindt?

De aspecten van communicatie - die zijn gevonden om deze contributing factors te analyseren - zijn de actoren, afstand, timing, stroom van informatie, vorm, zintuigen en communicatie media. Deze aspecten zijn gespecifieerd voor elke contributing factor die is gevonden, in de 31 ongeluk en incident onderzoeksrapporten van de DSB die geanalyseerd zijn. De analyse van de verzameling van deze contributing factors zou uitwijzen wat voor soort interacties het meest frequent voorkomen in de onderzoeksrapporten.

Interacties tussen luchtverkeersleiders en andere mensen in de luchtvaart operatie zijn vaak gevonden als contributing factor in de rapporten. Mens-

document interacties en mens-machine interacties zijn vaak gevonden, dit impliceert dat mensen in luchtvaart operatie problemen ondervinden met het begrijpen van de gereedschappen die ze gebruiken.

Een aantal aanbevelingen zijn gemaakt voor toekomstig onderzoek aan de Aviation Academy over dit onderwerp. In dit onderzoek is bevonden dat er een overgrote hoeveelheid informatie kan worden geproduceerd van alle resultaten. De aanbeveling is om het aantal aspecten te reduceren, dit zal namelijk het analyse proces versimpelen, een duidelijkere tool opleveren en meer handelbare resultaten geven. Een tweede aanbeveling die wordt gemaakt, is om de actoren in de interactie accurater te specificeren door afkortingen van de actroren te gebruiken, in plaats van alleen te beschrijven of de actor mens of niet mens is. Dit zal meer overzicht geven in resultaten en een accuratere post analyse opleveren.

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Introduction

1.1 Communication and communicational problems in aviation

Communication is an abundantly wide discussed subject among mass media, ethics, linguistics, bio communication, information theory, political sciences and a long list of other fields of study. Communication is described as an egocentric process (Epley, Keysar, Van Boven, & Gilovich, 2004), defined as the experience of self and others through dialogue (Chang, 1996) and theorized as the reproduction of social order (Kvasny, 2000).

Large scale disasters such as Bhopal disaster (Shrivastava, 1987), the Linate Airport disaster (Flottau, 2001) and off course Tenerife (Weick, 1990), have been linked to communicational problems. Research has found that in 70% to 80% of all aviation incident reports, human to human communicational problems are present (Billings & Cheaney, 1981; Sexton & Helmreich, 2000). These findings suggest that communicational problems amongst humans could play a large role in accidents and incidents that occur in aviation.

Catastrophes such as the Air Inter Flight 148 crash (Sparaco, 1994), the Air France 447 crash (Wise, 2011) and Chernobyl (Balonov, 2007) have been linked to human machine interaction issues. Although no simple percentage of these problems is found, 'automation' problems and ergonomics problems in general are commonly found in aviation accidents (Endsley, 1999; Inagaki, 2003; Pizziol, Tessier, & Dehais, 2014).

1.2 Assignment and background

This research is performed at the Research Department of the Aviation Academy in Amsterdam, a department that researches various aviation related subjects. The research was conducted from February 2016 until July 2016.

The assignment for this research (Appendix I: on page 59) was to create an analysis tool which can reveal which communication types, models and problems are mostly present in accident and incidents. This analysis tool would be based on

the different type of actors and kinds of communication aspects that are found in literature.

1.3 Goal and research question

The main goal of this research is to produce a viable tool which can be used to analyse contributing factors in accident and incident investigation reports. The method that has been assigned to perform this analysis, is to analyse and describe the type of interactions in the contributing factors, instead of describing their problems, which is done often in previous research (Billings & Cheaney, 1981; Bürki-Cohen, 1995; Cardosi, 1994; Corradini & Cacciari, 2002; Lingard et al., 2004; Rakas & Yang, 2007). These studies analysed interpersonal communication for its problems, specifying certain interactions as 'readback error', 'message not originated' or 'callsign confusion'. This research will specify contributing factors on their kind of interaction in different elements, or aspects. For example, a readback error between crew and ATC will be a human-human remote radio interaction. This method of analysis will give descriptive results on which interactions are often described as contributing factors, instead of which types of problems. This way of thinking in mind, should bring clarity to the research question:

What analysis tool can be used to analyse aviation accident and incident investigation reports, so to identify the kind of interactions that are most frequently stated as contributing factor in these reports, and thereby give insight in what types of communicational problems aviation operation experiences?

1.4 Sub-questions of this research

Sub-questions were formulated to help answer this main research question, the first two sub-questions will be answered and discussed in the theoretical framework:

- *How is communication defined and on which definition will this research be based upon?*
- *Which aspects of communication are found in communication sciences, whereby interactions can be dissected?*

The third research sub-question will be answered and discussed in the methodology:

- *How are the aspects of communication enclosed in a reliable analysis tool, which can be used to dissect contributing factors in aviation accident and incident investigation reports regarding communication?*

The last sub-question will be answered and discussed in the discussion:

- *Which kinds of interactions are most frequently found in contributing factors stated in aviation accident and incident investigation reports and what do these findings implicate regarding communicational problems in aviation operation?*

2 Theoretical framework

2.1 Literature about communicational problems in aviation

2.1.1 Introduction to communicational problems

Communicational problems are frequently found in intrapersonal contact, as humans often take an egocentric role as both sender and receiver (Keysar, 2007). As important information is transferred among people and machines in aviation, miscommunication can have large consequences. Human factors issues related to interpersonal communication have been found in approximately 70% to 80% of all accidents (Billings & Cheaney, 1981; Sexton & Helmreich, 2000). In 13% of aviation accidents, communication issues between pilots and controllers can be clearly defined as causal factor (Gero, 2013).

2.1.2 'Effective communication'

Standardized formats, terminology, phraseology and procedures have been implemented to reduce communicational problems in all parts of aviation (Orasanu & Davison, 1997). Sexton and Helmreich (2000) tested whether language use of cockpit crews has influence on the numbers of errors crews make. They found that the use of short words and the use of the word 'we' was negatively correlated with failures, suggesting that communication and performance improve with the use of short words and the use of 'we'. They further argue that the use of 'we' creates an atmosphere in which crew members can communicate more freely.

Effective communication has also been linked to shared mental models (Baxter, Besnard, & Riley, 2007; Converse, 1993). Mental models can be seen as interpretations of the world, if these models are shared by actors in a system, such as control towers, operating rooms or cockpits, the performance will be better (Baxter et al., 2007). A problem with mental models, however, is the (faulty) assumptions that come with them. If systems produce information which deviates from reality (e.g. pitot-tube freezes and wrong speed is displayed on PFD) - while it functions properly according to the pilots using it - this could lead to a mismatch

between real word and internal presentation of pilots. Pilots could overtrust outcomes of aircraft displays, which can lead to decisions being based on wrong data, influencing flight outcomes and performance negatively (Pritchett, Vándor, & Edwards, 2002).

2.1.3 Types of communicational problems

Clearance composition, phraseology, delivery technique and readback/hearback communication errors were frequently found in incident reports regarding pilot-air-traffic controller communications (Morrison & Wright, 1989). In miscommunications between pilots and controllers, it was found that clearances often included wrong heading or altitude assignments. Controllers were also found to differ from standard terminology, causing confusions. Controllers were found to fail often in correcting erroneous flight crew readbacks of their instructions (Morrison & Wright, 1989). 'Absent mindedness' is a frequently occurring phenomenon in communicational problems. It is a characteristic of highly skilled or habitual activities, so it is a problem only experts counter, not beginners (Mcmillan, 1998). Absent mindedness is, for example, an ATC controller giving an altitude for an approaching aircraft out of habit while an uncommon departure route is active, which could cause a loss of separation between two aircraft.

In a landmark study by Billings & Cheaney (1981), 10.815 incidents, that were reported between 1978 and 1979, were found to have information transfer problems, this was more than 70% of the reports. They found that in 38% of the cases, a message that was supposed to be sent did not originate. In 37% of the cases, the message was inaccurate, incomplete, ambiguous or garbled and in 13% the message was untimely. The researchers observed that many problems were caused by expectancy, the receiver assumed to hear a certain message. They found that in only 3% of the cases, equipment was to blame for the communicational problem. Although this study is of a notable date, the findings are still in line with what is found in later studies (Cardosi, 1994; Corradini & Cacciari, 2002; Sexton & Helmreich, 2000).

2.1.4 Human-machine interaction problems

Conflicts in communication do not only occur between humans, but also between a human operator and artificial systems (Hoc, Young, & Blosseville, 2009; Pizziol et al., 2014), this is known as 'automation surprises' (Sarter, Woods, & Billings, 1997). This means that a human does not expect a system to behave in certain kind of way, which inhibits safety. This gap of understanding between humans and machines, and the hazardous situations that can occur due to this gap, can be explained by careless system design (Inagaki, 2003) and the level of automation (Endsley, 1999). Pilots were found to be surprised by autopilot and auto throttle systems, even after thousands of hours of experience on the same aircraft (Sarter et al., 1997).

A problem found in visual displays is that often, information is presented in a complex and high density manner (van Veen & van Erp, 2001). Visual information on displays is usually restricted to single locations, thus immobile and this makes it hard to notice changes, especially with various visual information sources present at the same time (van Veen & van Erp, 2001) .

2.2 Defining communication

2.2.1 Sub-question one

This part of the report will give answer to the first sub-question:

How is communication defined and on which definition will this research be based upon?

2.2.2 Why define communication?

Communication is an extremely wide phenomenon. Communication theorists agree the term is "seriously overburdened" and no single model can explain communication as it is (Carey, 1982; Clevenger, 1991; Dance, 1970; Deetz & Putnam, 2001; Slack, 2005; Taylor, 1992). The term communication is widely used and all fields of communication studies operate on various differing and sometimes contradicting definitions. For this reason, it is of importance to the research that the term 'communication' is defined, as this will also give a definition to what is meant with 'communicational problems'. A definition should be evaluated on how

well it helps to answer the questions that are investigated in the research (Littlejohn & Foss, 2008).

2.2.3 Defining communication by choosing an approach

This research focuses on what communicational problems aviation operation experiences, in a practical, systematic manner. A system is defined as "a group of devices or artificial objects or an organization forming a network especially for distributing something or serving a common purpose" (Merriam-Webster, 2016). For the purpose of this research, it is proposed to look at aviation operation as a system, with separate parts that communicate with each other with the common purpose being aviation safety. This presumption helps us to define communication, as it narrows our view on communication.

In his landmark article, Craig (1999) proposed 'the seven traditions of communication theory', in his attempt to unify the academic field of communication theory. The seven traditions of Craig (1999), will be used to determine in which tradition this research will operate in and will give a working definition for the term 'communication'.

Craig's (1999) seven traditions of communication theory (Appendix I: 59) is composed of the rhetorical, semiotic, phenomenological, cybernetic, socio-psychological, sociocultural and critical approach. The cybernetic approach is by far the most appropriate for our research, as it theorizes communication as information processing/transfer and takes a systemic view on communication.

2.2.4 Conclusion sub-question one

To answer the sub-question of this part of the report:

How is communication defined and on which definition will this research be based upon?

Communication is theorized as a system in this research, communication is defined in this research as 'information processing/transfer between actors'. Consequently, communication problems are situations where information was not processed or transferred properly.

2.3 Aviation operation as a communication system

This paragraph is dedicated to clarify the 'systemic view' on aviation operation in this research. This paragraph also takes a bird's eye view on communicational processes, which gives the reader insight on how interactions are dissected. Figure 2.3-1 is a schematic view of different actors in aviation that interact with each other in different ways, times and places. The figure should not be seen as an all-inclusive schematic of aviation operation, but rather an example with limited parts which is used to visualize the text below.

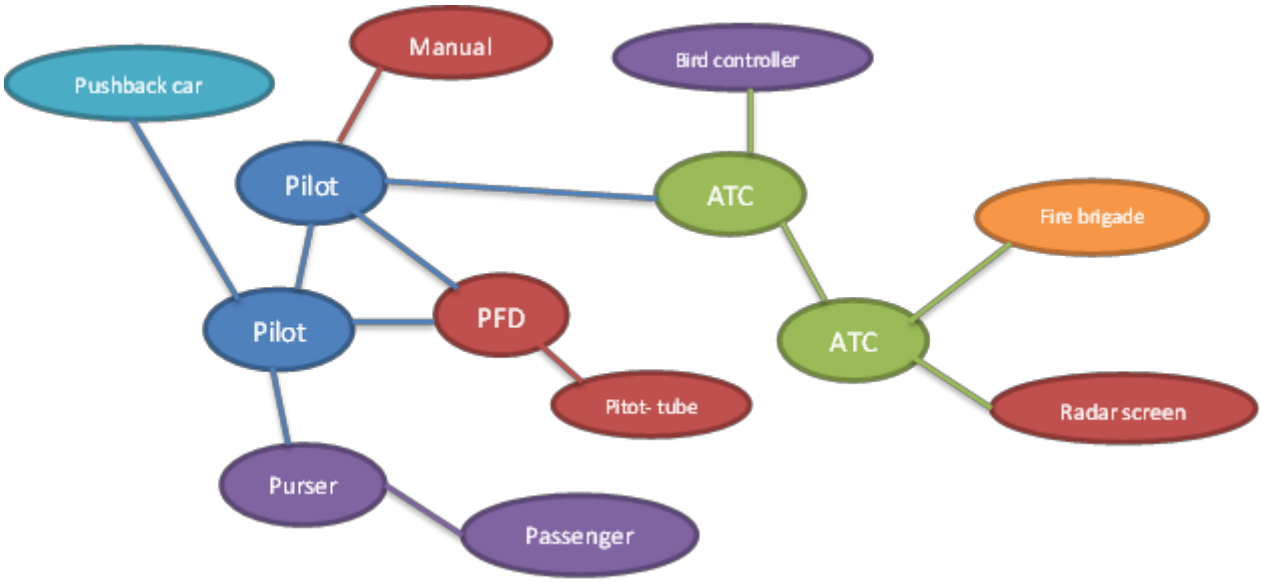

Figure 2.3-1: Example of aviation operation as a communication system

All these actors supply each other of information for the purpose of safety; the pilot reads the airspeed from the primary flight display (PFD), pushback car worker confirms the pushback car is clear from the aircraft to the crew, ATC clears the bird controller to check a runway, the purses gives passengers a safety briefing etc.

These interactions, interaction being a connection between a sender and receiver, all differ in the way the information was processed and transferred. The pilot retrieves the information from the PFD, pushback personnel waves to confirm that the car is clear of the aircraft, ATC clears the bird controller via radio transmission and the purser tells the passengers what to do in case of an emergency. These interactions have different forms, for example, in some interactions the actors are close to each other and in some they need a device for information transfer, such as a radio or a phone. The timing can also differ, for

example, a pilot receives a landing clearance from ATC right away and a manual is written long before it is read. These aspects of communication are elaborated in the next paragraph.

2.4 Defining the aspects of communication

Seven *aspects* of communication are found in literature, which can be used to describe interactions between actors in a communication system. The following aspects were found: who the actors are, what the distance is between them, the timing of the information transfer, the flow of information, the form of the information transmitted, the senses involved and the communicational media used in the interaction. Each aspect can be specified in different ways (e.g. in intra-crew communication the distance is local), these specifications are defined in this part of the report as well.

2.4.1 Sub-question two

This part of the report will give answer to the second sub-question:

Which aspects of communication are found in communication sciences, whereby human – human, human – machine and machine – machine interactions can be dissected?

2.4.2 Actors in interaction

An actor is defined as a sender or receiver of information. Aviation operation has a variety of actors, varying from checklists to pilots to navigation screens to bird controllers. As virtually every actor in aviation operation is either a human or is produced by a human, a distinction must be made between human actors and machine/technical actors. This distinction will be made on what actors directly interact with each other. With these definitions in place, all interactions in aviation operation can be categorized as a human-human, human-technical (machines, computers, manuals) or machine-machine interaction.

2.4.2.a Human actors

Human actors are, for example pilots, air traffic controllers, bird controllers, firemen, cabin crew, airport marshals. These are people that interact with other people or machines in the aviation system.

2.4.2.b Technical or machine actors

Machine actors are, for example, navigation screens, pitot tubes, air data computers, ailerons, radar screens, radar towers. An elaboration is needed on the term 'technical actors'. Manuals, taxiway lines, checklists, signs - much the same as displays and screens in the cockpit - are used by humans as tools for safe operation. These actors can thus be seen as 'technical' actors, which would fall under the category machines. The term 'machines' would be ambiguous, when in fact discussing manuals, paint or infrastructure, for this reason, the term 'technical' was created.

Machine – machine interactions will remain defined as such. Since human interactions with machines and human interactions with the 'technical' actors as discussed before are very much alike, from now on human-machine interactions will also be placed under the term human – technical.

2.4.3 Distance between actors

The distance between two actors can also be specified. An interaction can be either local or remote (Schulze, 1997). A differentiation in definition is made between human – human and human-technical interactions, and machine – machine interactions. For this reason, the subparagraphs are divided in the kind of actors involved in the interactions.

2.4.3.a Distance in human – human and human – technical interactions

For human – human and human – technical interactions, the interactions are **local** if the actors share the same space. In **human – human** interactions, examples for local contact are: a captain and co-pilot in cockpit, controller and supervisor conversing in ATC tower, purser and passenger in the cabin. In **human – technical** interactions, examples for local contact are: a controller and radar screen in the ATC room, a captain reading operating manual.

For human – human and human – technical interactions, the interactions are **remote** if the actors do not share the same space. In **human – human** interactions, examples for remote contact are: a captain and purser talking via phone, bird controller and ATC talking via radio, co-pilot confirming he has seen the airport

marschaller wave to him. In **human – technical** interactions, examples for remote contact are: a bird controller following taxiway lines while driving, controller visually confirming the position of a tug car, crew listening to ATIS message on radio.

2.4.3.b Distance in machine – machine interactions

For machine – machine interactions, the interaction is **local** if the machines are physically connected by cables, actuators etc. For example: an air data computer and primary flight display, the yoke and the aileron by mechanical actuator, autopilot and the flight management system.

For machine – machine interactions, the interaction is **remote** if the machines are not physically connected, but via radio, phone, internet etc. For example: the flight radio and ATC radio, ILS and Localizer receiver in the aircraft, ACARS and the ground station.

2.4.4 Timing of communication

The timing of the communication can also be specified for interactions. The timing can be divided in two ways, in synchronous and asynchronous (Schulze, 1997).

2.4.4.a Synchronous

Synchronous means 'happening, existing or arising at precisely the same time' (Merriam-Webster, 2016). This means that the receiver processes the message from the sender at practically the same time at which the message was produced; there is no or hardly any delay. Examples for this are: a phone call, radio contact, reading an aircraft's altitude on the PFD.

2.4.4.b Asynchronous

Asynchronous means 'not synchronous' (Merriam-Webster, 2016), which constitutes to not happening, existing or arising at the same time, but rather communication that includes a delay. The receiver does not process the message from the sender at the time it is produced, but the message is stored before it is processed. Examples for this are: email, voice mail, manuals, ATIS.

2.4.5 Information flow

Information flow is in what manner information is transferred between two actors, is the information flow both ways, or is it one-way. Heldman (2007) defined three different types of flow of information, namely: interactive, push and pull.

2.4.5.a Interactive information flow

Interactive means "mutually or reciprocally active" (Merriam-Webster, 2016). In communication this means that two, or more, actors both function as sender and receiver. Examples for this are: conversation, a phone call, the FMS and AP continually exchanging information.

2.4.5.b Push information flow

Push communication is a one-way method of sending information, defined as "stimulation to activity" (Merriam-Webster, 2016). In a communication system this means one actor sends information and the receiver is not expected to give a notable response. Examples for this are: sending a memo, input of data in autopilot, voicemail.

2.4.5.c Pull information flow

Pull communication is a one-way method of retrieving information, defined as "a force drawing someone or something in a particular direction" (Merriam-Webster, 2016). In communication, this constitutes to the retrieval of information from somewhere. Examples for this are: reading a book, seeing the altitude on the PFD, reading a checklist.

Pull communication is not considered in human-human interactions in this research, as a question is needed to retrieve information from one human actor to another. The retrieval of, for example, nonverbal cues in human-human interactions is thus defined as interactive.

2.4.6 Form of information

As this research focuses on communication between any actors in the aviation operation, be it human or machine, verbal and nonverbal communication theory will be generalized for practical purposes to human-technical and machine-machine interactions. A distinction is made between verbal and nonverbal communication;

verbal being the information transfer via symbols, including words, signs and numbers, and nonverbal being information transfer by intonation, bodily gestures, postures, physical force etc.

2.4.6.a Verbal communication

Verbal is commonly defined as "relating to, or consisting of words" (Merriam Webster, 2015). Verbal communication is defined in this research as information transmission via the meaning of symbols; the use of words, signs and numbers. Computers which communicate with unique symbols, such as the attitude indicator and navigation screens, are seen as verbal communication. Verbal communication can be spoken, written or coded language, such as binary or Morse code.

In human-human interactions, verbal communication constitutes to talking or writing between two human actors. In human-technical interactions, verbal constitutes to the display or the input of information by symbols (e.g. a knob is turned to enter a **value**). In machine-machine interactions, verbal constitutes to machines digitally communicating.

2.4.6.b Nonverbal communication

Nonverbal is commonly defined as "being other than verbal" (Merriam Webster, 2015). Ambady & Rosenthal (1998) defined nonverbal communication as "communication and interpretation of information by any means **other than language**". Behavioural and expressive channels of communication, such as bodily movements, physical force, vocal intonation and pitch, facial expression are seen as nonverbal communication channels (Knapp & Hall, 2009).

In human-human interactions, nonverbal communication constitutes to, for example, physical contact, waving, winking and screaming. In human-technical interactions, nonverbal constitutes to the display of information in the form other than a symbol or input of information via force (e.g. yoke push). In machine-machine interactions, nonverbal constitutes to machines analogically communicating (e.g. actuator deflecting aileron).

2.4.7 Senses involved

Aristotle (350 B.C.) is publicly thought to have first proposed the idea of five senses: touch, sight, hearing, smell and taste. Multiple senses can be involved in an interaction, for example: seeing someone, vocally greeting and consequently giving a hug. As the definition of the senses is considered to be self-explanatory and going deeper in this subject would only obscure things, no further attention will be given to the subject in question.

2.4.8 Communicational media

A communicational medium is a tool that is used to communicate between two or more actors. There is a great amount of communicational media in aviation operation. Pilots and controllers talking via radio communications, controllers with airport personnel via radio and phone, crews and airlines with internet etc. A variety of combinations can be made. An all-inclusive list of communicational media used in aviation operation could not be found in literature. For this reason, a small explorative study in incident reports was performed by Harald Luiks, the graduate researcher, which yielded the following communicational media occurring often:

- **Radio** was found commonly, as many interactions in aviation operation are performed via this medium.
- **Document** was also found in reports, as checklists and manuals are often used.
- **Displays and devices** were found often, for the cockpit and control tower are full of these.
- **Infrastructure** was found in events were pilots, for example, misunderstood taxiway markings.
- **Phone** communications were found, in both operational as non-operational interactions.
- **Aural warnings** were found a few times in the form of TCAS and EGPWS warnings.
- **Cable / actuators** were found once, this communicational media is specific for machine – machine interactions.

- **Internet** was not found, but due to its high applicability in the world per se, it is deemed relevant.

2.4.9 Conclusion sub-question two

To conclude this part of the report, the subsequent sub-question is answered:

Which aspects of communication are found in communication sciences, whereby human – human, human – machine and machine – machine interactions can be dissected?

The aspects that were found are enclosed in Table 2.4-1 below:

Aspect	Explained	Different specifications
Actors	The actors that are involved in the interaction	Human-human, human-technical, machine-machine
Distance	The distance between the actors of the interaction	Local, remote
Timing	The timing between when the information was produced and when it was processed	Synchronous, asynchronous
Flow	The flow of information between sender and receiver	Interactive, push, pull
Form	The form of the information that is sent and received	Verbal, nonverbal
Senses	The senses that are involved in the interaction	Sight, hearing, touch, smell, taste
Communicational media	The communicational media that is used between the sender and receiver	Radio, document, displays/devices, infrastructure, phone, aural, cable/actuator, internet

Table 2.4-1: Aspects with respective descriptions and specifications

3 Methodology

This part of the report will elaborate on how the research was performed. The sample that is analysed will be discussed first, after which the methodology will give answer to the third sub-question.

3.1 The analysed sample

The sample used is secondary data of incidents and accidents, namely investigation reports by the DSB. The sample that is analysed is a report set consisting of 36 reports from the Dutch Safety Board (DSB). This report set was supplied to the researcher by the Research Department and it is used in various other researches performed at the department.

DSB was chosen as investigation report source, for DSB is the Dutch safety investigation organisation that handles aviation accidents and incidents that occurred in the Netherlands or that originate from the Netherlands, such as the MH17 crash in Ukraine in 2014.

As this research is performed in the Netherlands, a preference is present for reports discussing incidents and accidents in the Netherlands. Other accident/incident investigation organisations, such as the American NTSB, the Canadian BFU, the British AAIB, are considered less suitable, for they investigate cases that occur in other parts of the world than the Netherlands.

3.2 Sub-question three

This part of the report will give answer to the third sub-question:

How are the aspects of communication enclosed in a reliable analysis tool, which can be used to dissect contributing factors in aviation accident and incident investigation reports regarding communication?

3.3 From theoretical framework to analysis tool

A tool was produced to analyse the contributing factors regarding communication found in investigation reports. This was done in such a way that the

interaction in the contributing factors is described, rather than the problem. The tool was made in the form of an excel sheet. The aspects that were found and described in the theoretical framework and their definitions are enclosed in the tool (Appendix III: on page 61).

In Appendix IV: 64, the practical part of the tool is found, this part of the tool is used for describing each contributing factor found in a report. The tool is build up in the following way:

- The seven aspects are divided in seven columns and each aspect is divided in its subsequent specifications.
- Information about the contributing factor is documented, by including the report number and a summary of the contributing factor, found in the report.
- There are two rows for each contributing factor: the blue row is used for statistical purposes and the grey row for notes.
    - A '1' is inserted in the cell below the specification that describes the contributing factor (e.g. human-human). The '1' is inserted for statistical purposes, this could also have been an 'x' or a '7'.
    - The grey row below the blue row is used for notes on why the aspect was described in a certain way.

3.4 Method of analysis

The tool works as follows: a contributing factor regarding communication is found in a report and each of the seven aspects is consequently specified (tables below or Appendix IV: on page 64). For example, this contributing factor was found in one of the reports:

'The crew of the Airbus A319 replied to the permission to take off intended for the Airbus A330'.

The tool is filled from left to right in the following manner (Table 3.4-1 on next page):

		What/who are the sender(s) and receiver(s) in the contributing factor?			Are the actors in the same space (**local**) or do they communicate remotely (**remote**)?		Does the receiver receive and notice the message right away (**synchronous**) or is there a delay between message produced and message received (**asynchronous**)?	
Report	Factors	Human-Human (HH)	Human-Technical (HT)	Machine-Machine (MM)	Local	Remote	Synchronous	Asynchronous
2007015	Factor 1	1				1	1	
Notes->	The crew of the Airbus A319 replied to the permission to take off intended for the Airbus A330.	ATC-A319: The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances.				The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances	The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances	

Table 3.4-1 : Example of a contributing factor specified

- The reference number of the DSB is inserted in the first column, for documentation purposes.
- As multiple contributing factors can be present per report, a number per factor is documented next to the report number.
- The contributing factor in question is human-human, as the interaction is between ATC and a crew of an Airbus A319. For this reason, a '1' is inserted in the cell below 'HH' and proof is given.
- As the ATC and the crew interact with each other via radio, the interaction is specified as 'remote', a '1' is inserted in the proper cell and proof is given.
- For the crew received the message from ATC directly, the interaction is specified as 'synchronous'.
- As this interaction is between a crew and ATC consists of both the crew and ATC sending and receiving messages, the interaction is specified as 'interactive' (see table on next page).
- For the interaction is of verbal nature, as the crew replied to a permission that was broadcasted by ATC, the interaction is specified 'verbal'.
- The only sense that is deemed relevant in this interaction is 'hearing', as the interaction was of aural nature only.

Do the actors both send messages actively (**interactive**), does only one actor send and the other actor receives (**push**) or does one actor retrieve information on own initiative (**pull is applicable to human-human**)?			Is the information transmission in the contributing factor via language; words, numbers or code (**verbal**) or is it via lights, visuals, vocal tone, physical force, bodily movements etc (**nonverbal**)?		Through which sense(s) is the information perceived by the receiver of the information? (Leave the cells null in case of machine-machine interaction)				
Interactive	Push	Pull (not for HH)	Verbal	Non-Verbal	Sight	Hearing	Taste	Smell	Touch
1			1		1				
The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances.			The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances.		The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances.				

Table 3.4-2: Example of a contributing factor specified (continued)

- Finally, the interaction is specified 'radio', for this is the communicational media used for information transmission in this interaction (see next table).

What is the communicational media between the two actors in the contributing factor? (not applicable to **local human-human** interaction)								
Radio	Phone	Internet	HT Display /Device	Infrastructure	Document	HT aural warning	Cable or actuator	Remarks
1								
The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances								

Table 3.4-3: Example of a contributing factor specified (continued)

- If certain aspects could not be specified or if a communicational media was used, which was not in the set list, a remark can be made in the remarks section.

3.5 Performing an inter-rater reliability test

An inter-rater reliability test was performed to find out if the tool is replicable and reliable. An inter-rater reliability test is a test where multiple people use the same tool to analyse the same sample of reports. The results of the analysis are

compared by the use of statistics, to find out if the results are statistically far apart or close together. The inter-rater reliability was calculated with the 'Intraclass Correlation' calculation in SPSS. Inter-rater results with at least 75% agreement are deemed acceptable. An inter rater reliability test was performed twice, because a first attempt failed.

In the first attempt, four people analysed five incident reports from the Aviation Safety Reporting System database (ASRS). ASRS reports are short incident reports made on own initiative by reporters, this report set was first considered to be used for this research. This attempt failed, as the results were 64.1% in agreement (Appendix V: on page 67). The raters indicated that this might be due to unclear definitions and ambiguous reports.

After this failed attempt, aspect definitions were reviewed and clarified. This failed attempt was also cause to choose a different sample set, the DSB report set supplied by the Research Department.

The second attempt, three people analysed three investigation reports from the DSB report set (Appendix VIII: 73). Three short, approximately five paged, DSB reports were chosen as sample set. Asking three people to read through and analyse three 60 paged accident investigation reports without commission, was considered 'too much to ask'. The agreement between the raters was 98.2%, thus the agreement was deemed acceptable and the tool was proven to be replicable and reliable.

3.6 Processing the analysis results

After the analysis of the reports, the results are processed to make them viable for statistical calculations and to make them clearer and more concise for the eye.

3.6.1 Initial processing of results

The results of the analysis are given a first screening. In the event that it is found that certain specifications, such as the sense smell, are not specified, then the columns belonging to these specifications are removed from the results. This action will bring forth simpler, more concise data, without compromising the quality of the data.

3.6.2 Coding the analyses results

After the first screening, all the blue rows are retrieved from the analysed contributing factors, containing the 1's in the columns describing the specifications of each contributing factor. These 1's are substituted to the subsequent abbreviation for each specification, for example: a 1 specifying the distance is local, will be replaced by 'LOC'. After all these 1's are substituted for their abbreviations, the abbreviations are moved to a single column per aspect. As multiple senses can be specified per contributing factor, each sense is taken separately.

#DSB	#F	ACTORS	DISTANCE	TIMING	FLOW	FORM	Sight	Taste	Smell	Hearing	Touch	COMM
2003071	1	HT	LOC	SYNC	PULL	VERB	YES	NO	NO	NO	NO	DOCU

Table 3.6-1 Example of coded analysis results

The result (Table 3.6-1) is that the data is described in a manner which can be used in statistical analysis and is simple to understand. In Table 3.6-1 for example, from left to right, a **human-technical** interaction of **local** distance with **synchronous** timing, **pull** information flow, **verbal** form of information, **sight** only sense involved and **document** as communicational media, is described.

From this coded data alone, in Excel, frequencies of certain specifications can be determined. For example, how many times local distance was cited or how many times sight was described in contributing factor analysis. Furthermore, the frequency of certain exact specifications of contributing factors can be determined. For example, it can be seen how many times contributing factors were described as the human-document specification from the table above, being HT, LOC, SYNC, PULL, VERB, SIGHT, DOCU.

3.6.3 Performing statistical tests

3.6.3.a Introduction to Pearson's chi-squared test

A Chi Square test is a statistical test on independence. The test is performed to find out if there is a significant relationship between two aspects. This constitutes to one aspect being more or less frequently present when another aspect is. For example, the combination of the aspects 'actor' and 'timing': 'remote' could be specified significantly more frequent in contributing factors when 'human-human'

is specified and 'local' could be specified significantly more frequent when 'human-technical' is specified.

3.6.3.b Handling of test outcomes

The coded data is inserted in a datasheet in SPSS, a computer programme used for statistical analysis. Chi Square statistical tests are performed between all the aspects – in total 36 different combinations - the syntaxes for these calculations are enclosed in the appendices (Appendix VI: 68). The data is inserted in a way that all the aspects (actors, distance, timing etc.) are variables and the specifications (local, verbal, radio etc.) are nominal values belonging to each aspect.

The results of these calculations are in the form of two tables. One dedicated to the frequencies found in the data set for the aspects and one for the results of statistical tests performed. For a readable example would take up a whole page, an example of both of these tables is enclosed in Appendix V: on page 67.

The outcomes of Pearson's chi-squared test and other tests are elaborated. The outcome of such a statistical test, is a P-value. If this P-value is less than '0,05', the test data is declared dependent. This means that the frequency distribution of the data most probably did not arise by chance, but rather because the data - in our case the aspects - are dependent of each other. One aspect is significantly more or less frequently present, when another one is. A confidence level of 95% is chosen, which constitutes to the P-value limit of '0,05'.

The Pearson's chi-squared test has an assumption, or condition, that needs to be met for the test to work properly. This assumption is that no more than 20% of the expected counts are less than '5' and the minimum expected count is '1' or greater. This practically means that the data cannot contain more than a few low values. If there are more than 20% expected counts that are less than '5' or the minimum expected count is less than '1', other statistical test results are used. This test is the Fisher's Exact test, which works for low quantities as well. For this test, the same 95% confidence level is used, so if the P-value of this test is lower than '0,05', the data is declared dependent.

Table 2.4-1 on the next page, is an example of the outcomes of the statistical tests performed by SPSS. The P-value of the Pearson Chi Square test is '0,092' (blue circle). The Fisher's Exact test outcome is '0,146' (green circle). The data regarding the expected counts is included as well, stating that '25%' of the expected counts are less than 5 and the minimum expected count is '4,9' (red circle). It was stated earlier, that the Fisher's Exact test would be used, if the expected counts more than 5 exceeded 20%, which is the case in this example. As the Fisher's Exact test has an outcome of '0,146' in this example, the data is not significantly dependent.

	Value	df	Asymptotic Significance (2-sided)	Exact Sig.(2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	2,831[a]	1	0,092 1		
Continuity Correction[b]	1,939	1	0,164		
Likelihood Ratio	3,282	1	0,07		
Fisher's Exact Test				0,146 2	0,076
N of Valid Cases	102				

a. 1 cells (25,0%) have expected count less than 5. The minimum expected count is 4,90. 3

b. Computed only for a 2x2 table

Table 3.6-2: Example of Chi Square results

3.6.3.c Data handling

In the event that some values or variables are present in a very low quantity, which would bring forth data that is insufficient for statistical analysis, then these values or variables are either changed to 'none specified' or are removed all together, as the contributing factor can then be seen as an outlier. This would need to be considered per case.

3.7 Conclusion sub-question three

To conclude this part of the report, the subsequent sub-question is answered:

How are the aspects of communication enclosed in a reliable analysis tool, which can be used to dissect contributing factors in aviation accident and incident investigation reports regarding communication?

The aspects that were described in the theoretical framework are enclosed in tool in the form of an excel sheet. The aspects are described per contributing factor in a structured, consecutive manner. The tool was proven to be reliable through an inter-rater reliability test.

4 Results

4.1 Familiarisation of the results

Of the 36 reports in the report set, 31 were analysed with the analysis tool. Five of the reports did not discuss communicational problems. For this research is focused on *types* of communicational problems and not the frequency of communicational problems per se, these reports were not analysed nor enclosed in the results. The analysis of the 31 reports yielded 103 contributing factors relating to communication, varying from crew resource management problems to ambiguous clearances from ATC to auto-throttle automation problems.

4.2 General frequencies

For clarity and aesthetic purposes, the general frequencies of each aspect are divided in two tables.

	Actors			Distance		Timing		Information flow		
	Human-human	Human-Technical	Machine-machine	Local	Remote	Synchronous	Asynchronous	Interactive	Push	Pull
Frequency [N]	40	62	1	55	48	78	25	45	45	13
Frequency [%]	38,83	60,19	0,97	53,40	46,60	75,73	24,27	43,69	43,69	12,62

Table 4.2-1: Frequencies of aspects

Table 4.2-1 above presents the frequencies found for the different *actors* in the contributing factors, *distance* between these actors, *timing* of the information transfer and the *flow* of *information*. Of the 103 contributing factors, 62 were categorized as 'human – technical' (60.2%), 40 as 'human – human' (38.8%) and only one as 'machine – machine' (0.97%). Regarding distance, 55 contributing factors were specified as 'local' (53.4%) and 48 as 'remote' (46.6%). The timing of the contributing factors was specified 'synchronous' 78 times (75.7%) and 'asynchronous' 25 times (24.3%). The information flow of the factors was specified as 'interactive' 45 times (43.7%), as 'push' 45 times (43.7%) and 13 times as 'pull' (12.6%).

	Verbal		Senses					Media						
	Verbal	Nonverbal	Sight	Hearing	Smell	Taste	Touch	Radio	Document	HT Display / Device	Infrastructure	HT Aural Warning	Cable	None specified
Frequency [N]	83	20	49	50	0	0	11	32	16	25	3	1	1	25
Frequency [%]	80,58	19,42	47,57	48,54	0,00	0,00	10,68	31,07	15,53	24,27	2,91	0,97	0,97	24,27

Table 4.2-2: Frequencies of aspects (continued)

Table 4.2-2 above presents the frequencies of how many times a contributing factor was specified as *verbal* or *nonverbal*, which *senses* were involved and which *communicational media* was used in the information transfer. The contributing factors were specified as 'verbal' 83 times (80.6%) and 'nonverbal' 20 times (19.4%). The sense 'sight' was specified 49 times (47,6%), 'hearing' 50 times (48.5%) and 'touch' 11 times (10.7%), the other senses were never specified. The percentages for the senses add up to more than 100%, because in some cases more senses were specified. 'Radio' was specified 32 times (31.1%), 'document' 16 times (15.5%), HT display/device 25 times (24.3), infrastructure 3 times (2.91%), HT Aural Warning and cable one time (0.97%). In 25 contributing factors, no communicational media was specified (24.3%).

4.3 Relationships between aspects

4.3.1 Omitted data

Some specifications were only present in a very low quantity; these were removed, because these quantities are insufficient for a statistical analysis. The variables that were removed are 'smell' and 'taste', as these were not found. Some of the communicational media, these were 'Infrastructure', 'HT Aural Warning' and 'Cable', were found only a few times, these were changed to 'none specified'. The one machine - machine contributing factor that was found, was removed altogether from the dataset for the statistical tests.

Specification	Frequency [N]	Changed to
HT Aural Warning	1	None specified
Infrastructure	3	None specified
Cable	1	None specified
Smell	0	Omitted

Taste	0	Omitted

Table 4.3-1: Omitted or changed data in dataset for statistical analysis

4.3.2 Relationships between aspects

Table 4.3-2 below shows the combinations of aspects and the consequent outcomes of the Chi Square analysis (the P values in the table). Statistically significant relationships, combinations of aspects with a P value under '0.05', were found in 26 out of the 36 combinations, six combinations were insignificant (blue numbers) and for four combinations it could not be calculated due to insufficient data (*). For the purpose of structure and overview, the elaboration of the 36 combinations is enclosed in Appendix XIII: on page 84.

	Actors	Distance	Timing	Info. flow	Verbal	Sight	Hearing	Touch	Comm. media
Actors		0,000	0,000	0,000	0,003	0,000	0,000	0,003	0,000
Distance			0,724	0,018	0,481	0,001	0,000	0,001	0,000
Timing				0,000	0,000	0,146	0,000	0,286	0,000
Info. flow					0,118	0,000	0,000	*	*
Verbal						0,028	0,000	0,000	*
Sight							0,000	0,647	0,000
Hearing								0,001	0,000
Touch									*
Comm. media									

Table 4.3-2: Relationships between aspects

(*data insufficient for statistical analysis, blue is insignificant)

4.4 Frequency of specifications

A lot of combinations of aspects can be found and looked into; this was done in the last paragraph for one vs. one aspect combinations. In this paragraph, combinations between aspects are presented which show certain specifications that were found frequently in contributing factors. These are not results of statistical analysis, but frequencies of aspect combinations.

4.4.1 The frequencies of 'actors' by 'distance'

As the following subparagraphs build forth on the 'actors' and 'distance' combination, the frequencies for this combination will be shown first (Table 4.4-1).

Combination	Frequency [N]	Frequency [%]
HT and Local	42	40,78
HH and Remote	28	27,2
HT and Remote	20	19,42
HH and Local	12	11,65
MM and Local	1	0,97
MM and Remote	0	0,00

Table 4.4-1 : Frequencies of 'actors' by 'distance'

Local human-technical was most frequently found, 42 times (40.78%). This was followed by remote human-human, which was found 28 times (27.2%). Remote human-technical was found 20 times (19.42%). Local human-human was found 12 times (11.65%) and local machine-machine was found only once (0.97%).

4.4.2 The frequencies 'actors' by 'distance' by 'timing'

The aspect of timing is added to the 'actors' and 'distance' combination, thus the combination 'actors' by 'distance' by 'timing' is created. The frequencies in this combination are more dispersed that in the 'actors' by 'distance' combination.

Combination	Frequency [N]	Frequency [%]
HH, Remote, Synchronous	28	27,2
HT, Local, Synchronous	28	27,2
HT, Local, Asynchronous	14	13,59
HH, Local, Synchronous	12	11,65
HT, Remote, Asynchronous	11	10,68
HT, Remote, Synchronous	9	8,74
MM, Local, Synchronous	1	0,97

Table 4.4-2: Frequencies of 'actors' by 'distance' by 'timing'

As seen in Table 4.4-2 above, remote human-human synchronous and local human-technical synchronous were both found 28 times (27.2%). Local human-technical asynchronous was found 14 times (13.6%). Local human-human synchronous was found twelve times (11.7%). Remote human-technical asynchronous was eleven times (10.7%). Remote human-technical synchronous was found nine times (8.7%). Local machine-machine synchronous was found only once (0.97%). Five other combinations were omitted from this table, as they were not found (Appendix IX: on page 75)

4.4.3 The frequencies 'actors' by 'distance' by 'timing' by 'information flow'

When the aspect 'information flow' is added to the 'actors' and 'distance' and 'timing' combination, a few combinations are more concentrated than others.

Combination	Frequency [N]	Frequency [%]
HH, Remote, Synchronous, Interactive	26	25,24
HT, Local, Synchronous, Pull	14	13,59
HT, Local, Asynchronous, Pull	12	11,65
HT, Remote, Asynchronous, Pull	10	9,71
HH, Local, Synchronous, Interactive	9	8,74
HT, Local, Synchronous, Interactive	7	6,80
HT, Local, Synchronous, Push	7	6,80
HT, Remote, Synchronous, Pull	7	6,80
HT, Remote, Synchronous, Interactive	2	1,94
HH, Remote, Synchronous, Push	2	1,94
HH, Local, Synchronous, Pull	2	1,94

Table 4.4-3: Frequencies of 'actors' by 'distance' by 'timing' by 'information flow'

As seen in Table 4.4-3 above, remote human-human synchronous interactive was found 26 times (25.3%). Local human-technical synchronous pull was found 14 times (13.6%). Local human-technical asynchronous pull was found twelve (11.7%). Remote human-technical asynchronous pull was found ten times (9.71%). Local human-human synchronous interactive was found nine times (8.74%). Local human-technical synchronous interactive, local human-technical synchronous push and remote human-technical synchronous pull was found seven times (6.80%). Remote human-technical synchronous interactive, remote human-human remote synchronous push and local human-human synchronous pull were found two times (1.94%). The rest of the combinations possible were found only once or not at all, these are enclosed in Appendix X: on page 76.

4.4.4 Seven most frequent exact specifications

As most interactions were specified as verbal (81%) and covering every sense would not give particularly more insight, but would delude the results section: these two aspects are not separately discussed. Out of the 16128 specifications possible, 41 unique specifications were found in the 103 contributing factors (Appendix XI: on page 77). These are the seven most frequently found specifications.

#	Combination	Frequency [N]	Frequency [%]
1	HH, Remote, Synchronous, Interactive, Verbal, Hear, Radio	22	21,36
2	HT, Local, Asynchronous, Pull, Verbal, Sight, Document	10	9,71
3	HH, Local, Synchronous, Interactive, Verbal, Hear	7	6,80
4	HT, Local, Synchronous, Pull, Verbal, Sight, HT Device/Display	7	6,80
5	HT, Remote, Synchronous, Pull, Nonverbal, Sight	5	4,85
6	HT, Local, Synchronous, Interactive, Nonverbal, Sight, Touch, HT Device/Display	4	3,88
7	HT, Remote, Asynchronous, Pull, Verbal, Hear, Radio	3	2,91

Table 4.4-4: Seven most frequent specifications

For practical and paper-saving purposes, in this paragraph the specifications (Table 4.4-4) will be referenced to by their number (column one). In 22 cases, a contributing factor was specified as one (21.4%). The human - technical interaction two, was specified ten times (9.71%). In seven cases, three or four were specified (6.80%). In five cases, a contributing factor was specified as remote human - technical interaction five (4.85%). Four contributing factors, were specified as local human - technical interaction six (3.88%). In three cases, combination seven was specified (2.91%).

A table with each contributing factor found and their respective specification is found in Appendix XII: on page 78.

5 Discussion

5.1 Sub-question four

The discussion will answer the fourth and final sub-question:

Which kinds of interactions are most frequently found in contributing factors stated in aviation accident and incident investigation reports and what do these findings implicate regarding communicational problems in aviation operation?

5.2 Introduction to discussing the findings

In this chapter, certain suggestions are made. These suggestions, though, are hard to make, because no total image of the frequency of certain interactions is present. A total number of interactions between the different actors in the aviation system is unknown. This makes it hard to see, for example, if human-technical interaction problems occur relatively often **or** if human-technical interactions are just very common in general. This problem in unknown relativity is a recurring problem in this study.

5.3 General frequencies

Human-technical interactions (60.2%) were most frequently found to be contributing factors, followed by human-human interactions (38.8%). This is a surprising finding, as it suggests that humans might experience more communicational problems in interactions with the tools they use than with other humans. Local (53.4%) and remote (46.6%) information transfer was found about equally as often, this might be an indication that the distance between two actors does not play a large role in the effectiveness of the information transfer. Synchronous (75.7%) timing was found far more than asynchronous (24.3%) timing, suggesting that information transfer with a delay may be more reliable than direct information transfer. This may be due to the extra care that is taken in asynchronous information transfer. Interactive and push information flow were both found in 43.7% of the contributing factors, suggesting problems in communication are more often attributed to the sender of the information. Pull

information flow was only found in 12.6% of the contributing factors, suggesting information is more often successfully pulled opposed to pushed. This may be because in pull communication, the receiver of the information knows which information it wants and is prepared for that information.

The majority of the contributing factors are of verbal nature (80.6%), suggesting that language and symbols are more often a problem than nonverbal (19.4%) cues. Sight (47.6%) and hearing (48.5%) were found most often of all senses; this is no surprising finding as it is considered that most information in aviation operation is perceived through these senses, rather than smell and taste. Radio (31.0%) and HT device (24.3%) are found most often in the communicational media aspect. As radio is a communicational media which is used almost exclusively for human-human remote interactions, this comes to no surprise. The high frequency of HT device contributing factors is also not striking, as the whole cockpit and all its components and manuals are specified as such.

5.4 Relationships between aspects

As only 42 unique specifications were found, out of the 16128 specifications possible, the results are rather concentrated. For this reason, 26 out of the 36 combinations between aspects have a statistically significant relationship. Moreover, because of this high concentration in results, some combinations were found to be insufficient for statistical analysis.

Some interesting relationships were found. It was found that 70% of all human-human contributing factors, were remote, suggesting that face to face human interactions are rather reliable. Human-human contributing factors were found to have ‘hearing’ specified more (76.0%) and human-technical contributing factors were found to have ‘sight’ specified more (93.9%). In 62.5% of all remote interactions, radio was the communicational media specified, this indicates that radio is a heavily used medium for communication in aviation operation.

5.5 Unique specifications

A portion of the table from the results is copied here for the purpose of ease Table 5.5-1. In this part of the report, investigation reports are cited.

#	Combination	Frequency [N]	Frequency [%]
1	HH, Remote, Synchronous, Interactive, Verbal, Hear, Radio	22	21,36
2	HT, Local, Asynchronous, Pull, Verbal, Sight, Document	10	9,71
3	HH, Local, Synchronous, Interactive, Verbal, Hear	7	6,80
4	HT, Local, Synchronous, Pull, Verbal, Sight, HT Device/Display	7	6,80

Table 5.5-1 Four most frequent specifications

The specifications are again referred to by their respective numbers. In 21.4% of the contributing factors, specification one was described. When Appendix XII: on page 78 is reviewed (row 17 to 31), it is seen in the summary that in 19 of these cases the interaction was between pilots and air traffic controllers. A read through these contributing factors learn that controllers not following standard phraseology, a concept discussed by multiple researchers (Billings & Cheaney, 1981; Corradini & Cacciari, 2002; Rakas & Yang, 2007), is a recurring problem. An example:

- *A misinterpretation of a clearance from the ground controller, in the course of which no standard terminology had been applied, resulted in the crew of G-LEAA taxiing onto runway 22 without permission.*

In some cases, insufficient information, due to a difference in mental models and assumptions that information is already present (Converse, 1993), was found to be a contributing factor. An example:

- *Though air traffic control performed its task and was entitled to assume its instruction would be followed up, the Dutch Safety Board holds the view that the slipperiness on VS and P6 could have been reported more explicitly to the crew of EZY5112.*

Cardosi (1994) found that readback errors were often a problem in taxiway instructions. Consistent of his findings, readback errors in taxiway instructions were found a few times:

- *The crew was already on the point where they were told to hold short, so they thought the controller meant to hold short of the next runway: "To*

confirm the clearance with the ground controller, we repeated his command and since there was no correction or objection from his side we began taxiing across runway 22 towards the holding position for the next runway on taxiway G5."

- *After being pushed-back from gate D3 the ground controller cleared the aircraft to 'taxi runway 36C, to leave via A10 and taxi via the south'. When the intersection with taxiway Alpha was reached, the flight crew turned to the left instead of to the right.*

In the landmark study of Billings & Cheaney (1981), which is still often cited in communicational problem researches, 'message not originated' is found in 37% of all information transfer problems. However, of the 22 human-human radio contributing factors found, only two state this problem:

- *The ATC controller guiding the Airbus from the east, did not give the crew a final clearance for the approach to runway 18C. [..] Because a clearance was not given, the crew persisted on the last given heading and crossed centreline.*
- *The driver of [bird control car] 2 stopped the vehicle on the runway near the end of runway 24 to clear the remains of an animal from the runway. The controller stated that [bird control car] 2 did not report this on the frequency, something that is normally done.*

This finding suggests that this research might be out-dated. The finding above can also suggest that communicational problems decreased and that certain types of communicational problems became less frequent, because of the focus on standardization, CRM and procedures since this research (Orasanu & Davison, 1997).

Only seven contributing factors (6.80%) were specified as 'HH, Local, Synchronous, Interactive, Verbal, Hear', most of which were interactions among crews. Most these interactions involved a lack of conversation, which could be seen as a lack of effective CRM (Maurino, Reason, Johnston, & Lee, 1998). An example:

- *[…] the Co-Pilot – other than the decision to make a go-around after the first approach – failed to let the Captain know in a positive way that she felt uncomfortable.*

In 9.70% of the contributing factors, human document interactions were specified as contributing factor. This is striking, as previous studies about communicational problems were not found discussing this subject. Most of the contributing factors found in human document interactions revolved around the absence of a clear definition, some examples:

- *The term 'active runway' is not clearly defined in the Air Traffic Control Operations Manual.*
- *In this respect it should be noted that by not establishing a clear and definite crosswind limit in the [airline] Operations Manual a defence barrier against unsafe operations was lost.*

It comes to no surprise that a machine-machine contributing factor was found only once, as human factors play the most important causal role in aviation accidents (Wiegmann & Shappell, 2001). This one machine-machine factor is not discussed, as it is considered that a sample of one is not of relevant in this research.

In 6.80% of the contributing factors, a human-machine interaction where information was pulled was specified, namely 'HT, Local, Synchronous, Pull, Verbal, Sight, HT Device/Display'. Most of these interactions, the human actor missed essential information or did not have access to the information. Missing information was discussed by van Veen & van Erp (2001), they considered it was due to the immobility of visual information that makes it hard to notice changes. An example:

- *As the airspeed continued to drop, the aircraft's pitch attitude kept increasing. The crew failed to recognise the airspeed decay and the pitch increase until the moment the stick shaker was activated.*

5.6 Conclusion sub-question four

To conclude this part of the report, the fourth and final sub-question is answered:

Which kinds of interactions are most frequently found in contributing factors stated in aviation accident and incident investigation reports and what do these findings implicate regarding communicational problems in aviation operation?

In the analysis of the summaries of the reports, interactions between ATC controllers and other human actors via radio were most frequently found, which could indicate that human-human radio contact is prone to communicational problems. Furthermore, an interesting finding was that human document interaction was often found as contributing factor. Moreover, human-machine interactions were found, where the human actors had issues understanding the machines they were using, due to automation. The finding that human-document interactions and human-machine interactions were frequently found as contributing factor, would indicate that human actors in aviation operation have difficulty understanding the tools they use.

6 Conclusion

To start concluding, the main research question is recited:

What analysis tool can be used to analyse aviation accident and incident investigation reports, so to identify the kind of interactions that are most frequently stated as contributing factor in these reports, and thereby give insight in what types of communicational problems aviation operation experiences?

For clarity, the conclusion will be in the form of the sub-questions which are answered in a consecutive manner.

6.1 Sub-question one

How is communication defined and on which definition will this research be based upon?

Communication is theorized as a system in this research, communication is defined in this research as 'information processing/transfer between actors'. Thus, communication problems are situations where information was not processed or transferred properly.

6.2 Sub-question two

Which aspects of communication are found in communication sciences, whereby human – human, human – machine and machine – machine interactions can be dissected?

Aspects of communication that were found in the theoretical framework were: who the actors are, what the distance is between them, the timing of the information transfer, the flow of information, the form of the information transmitted, the senses involved and the communicational media used in the interaction.

6.3 Sub-question three

How are the aspects of communication enclosed in a reliable analysis tool, which can be used to dissect contributing factors in aviation accident and incident investigation reports regarding communication?

The aspects and their respective definitions were enclosed in a tool, in the form of an excel sheet. This tool is used to analyse and describe the contributing factors regarding communication, found in the investigation report. Each contributing factor is analysed and described by specifying each aspect that is enclosed in the tool. The tool was found to be reliable and replicable through the use of an inter-rater reliability test, which agreement was 98.2%.

6.4 Sub-question four

Which kinds of interactions are most frequently found in contributing factors stated in aviation accident and incident investigation reports and what do these findings implicate regarding communicational problems in aviation operation?

It was found that human to human radio (21.4%) and human to document (9.7%) interactions were frequently stated as contributing factor. That human to human radio interactions were often found did not come as a surprise, as radio is a prime communicational medium in aviation operation. That human to document interaction were as often stated as was found, did come as a surprise. It was found that these contributing factors were mostly accounted for ambiguousness in definitions and what could and could not be done. Moreover, it was found that frequently human display interactions were cited as contributing factor. These findings indicate that humans in aviation operation have problems with understanding the tools they use.

7 Recommendations

In the first section of this part of the report, recommendations are given to the aviation sector, which strive to improve safety in aviation. As the assignment was to create a viable analysis tool, recommendations will be made on how this can be further achieved by the Research Department, in future research.

7.1 Recommendations for the aviation sector

7.1.1 Standard terminology/phraseology

A considerable amount of contributing factors were found to be attributed to the use of nonstandard terminology by Dutch air traffic controllers. It is recommended to Luchtverkeersleiding Nederland (LVNL) that it should review their performance regarding the use of standard terminology. This will give more insight in how this problem originates and consequently how it is prevented.

7.1.2 Ask more questions

It was found that communicational problems could easily be prevented by a simple question. For example, if a person has already been briefed on the icing situation on taxiways or if a certain procedure is known. A holistic recommendation is made to the aviation sector: Ask more questions. In reports it was often found that people were unsure of something and did not ask to clarify. They assumed that what they thought was right and did not want to burden anyone by asking a question. This recommendation is a little more ambiguous, because this problem is harder to quantify than other problems, such as the use of standard phraseology. It is, however, considered that this recommendation could prevent problems in aviation, for this reason it is given anyway.

7.1.3 Define clear limits and tasks

It was found that in human document interactions in aviation, most problems were attributed to unclear, ambiguous or absent limits or tasks. A recommendation is made to the whole Dutch aviation sector (e.g. airlines, LVNL, Schiphol), that it

should review on how the limits and tasks in the manuals are perceived by the people who use these manuals.

This could be done by performing a random sample test with a survey or interview, testing how different people perceive different limits and tasks. This should give an indication on how the perception of these limits or tasks differs among the people they apply to.

7.2 Recommendations for Research Department

7.2.1 Using multiple tools

In the development of the tool and the analysis process was found that exceptions were needed in the tool and the definitions, because it was made so it could describe human-human, human-technical and machine-machine interactions. It is recommended to look into dividing the tool into three parts, a human-human, human-technical and machine-machine analysis tool. This should make the analysis tool considerably clearer and easier to use.

7.2.2 Review aspect relevance

In the analysis process was found that some parts of the tool were not used, such as the senses 'smell' and 'taste' and the communicational media 'phone' and 'internet'. These parts were thus redundant. It is recommended, that the elements of the tool should be reviewed for redundancy. This should simplify the tool and ease the analysis process.

7.2.3 Specify the actors

As it is of now, the tool does not specify exactly who the actors are in each interaction, it just specifies whether they are human or not. Specific insight in whom or what the actors are, is considered to be useful information to have at hand in post-analysis. This could be done by building the tool in a way that from the specification of a contributing factor it can be easily seen, with eyes and by excel, who the actors are. The following form is proposed:

Contributing factor	Actor 1	Actor 2	Other aspects such as distance and information flow ->
1	CO-P	CAPT	LOC, …
2	CAPT	MANU	… ,ASYNC, …
3	ATC	RADAR	… ,PULL, …

Table 7.2-1: Future prospects

The abbreviations (Table 7.2-1) would stand for co-pilot, captain, manual, ATC controller, radar screen. These actors are also reduced in groups as follows:

Group	Actor	Actor	Other actors in group ->
Cockpit crew	CO-P	CAPT	COCO
Cockpit paper	CHECKL	MANU	SOP
Cockpit devices	AUTOP	FMS	NAVI
Airport personnel	TUG	BIRD	DE-ICE

Table 7.2-2: Future prospects 2

This method would give much more practical results than the results generated in this research. An initial list of actors and abbreviations can be made from the results of this research (Appendix XII: on page 78).

7.2.4 Future research

A final recommendation is made regarding future research on the subject of analysing contributing factors, regarding information transfer in aviation. The use of the knowledge and experience described in this thesis should result in a clearer tool which gives more insightful results and data. It is recommended to build forth on this research, using the knowledge of the aspects of communication, the practical method of the analysis tool and the recommendation regarding the specifying of actors.

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Abbreviations

HH	Human-Human
HT	Human-Technical
MM	Machine-Machine
PFD	Primary Flight Display
FMS	Flight Management System
ATC	Air Traffic Control
ILS	Instrument Landing System
ACARS	Aircraft Communications Addressing And Reporting System
TCAS	Traffic Collision Avoidance System
EGPWS	Enhanced Ground Proximity Warning System
DSB	Dutch Safety Board
ASRS	Aviation Safety Reporting System

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Appendices

Appendix I: Assignment of this research

GRADUATION ASSIGNMENT OPPORTUNITY IN THE RESEARCH DEPARTMENT OF THE AVIATION ACADEMY

ACADEMIC YEAR: 2015-2016

RESEARCH AREA: HUMAN FACTORS

POINT OF CONTACT: Dr. Nektarios Karanikas, n.karanikas@hva.nl, 0621156287

SUBJECT: Types of communication problems revealed from aviation accident investigations

Starting from: Any time

Background: Effective communication has been identified as a critical success factor for all activities, and reciprocally, poor communication has been found as factor of accidents in all industry sectors. There have been studies that presented the frequency of communication problems that contributed to accidents, but those studies have not been updated with latest findings and technology advancements and they do not always refer to types of communication flaws based on the various actors, human and equipment, kind of communication (e.g., synchronous and asynchronous, push and pull communication) and the causal factors suggested by the STAMP model (e.g., missing, delayed, overlapping, misinterpreted).

Description of the assignment: The student(s) based on a comprehensive review of literature referred to communication types, models and deficiencies will develop an analysis framework and analyse a large sample of accident reports. The analysis will reveal which communication types, models and problems are mostly present in accident and incidents. More particularly, the student(s) will:

1. Review the various existing communication models and types.
2. Review research results regarding communication problems which contributed to accidents and incidents.
3. Develop an analysis framework based on the review of step 1 above.
4. Pilot the application of the framework on a small sample of investigation reports and calculate the inter-rater reliability.
5. Refine the analysis framework based on the results of step 4 above and conduct further calculations of inter-rater reliability (iterative process).
6. Apply the framework on a large sample of accident reports.
7. Analyse the data, discuss the results and make further recommendations.

Research resources: The aforementioned research can be conducted by more than one graduate students; the more the number of students, the more the sample to be analysed and the stronger the results to be produced. In case that more than 1 student apply for this graduation assignment, the following terms apply:

- The maximum number of students is limited to three (3).
- The students will work as a team until they finalise the tool (steps 1 to 5 of the description above).
- Afterwards, each student will work independently (steps 6 and 7 of the description).
- Each student will submit an individual report, including his/her reflection and contribution to the steps 1 to 7 of the assignment's description.
- The grading of the thesis and presentation will be also individual.

Appendix II: The seven traditions of communication

	Rhetorical	Semiotic	Phenomenological	Cybernetic	Sociopsychological	Sociocultural	Critical
Communication theorized as:	The practical art of discourse	Intersubjective mediation by signs	Experience of otherness; dialogue	Information processing	Expression, interaction, & influence	(Re)production of social order	Discursive reflection
Problems of communication theorized as:	Social exigency requiring collective deliberation and judgment	Misunderstanding or gap between subjective viewpoints	Absence of, or failure to sustain, authentic human relationship	Noise; overload; underload; a malfuction or "bug" in a system	Situation requiring manipulation of causes of behavior to achieve specified outcomes	Conflict; alienation; misalignment; failure of coordination	Hegemonic ideology; systematically distorted speech situation
Metadiscursive vocabulary such as:	Art, method, communicator, audience, strategy, commonplace, logic, emotion	Sign, symbol, icon, index, meaning, referent, code, language, medium, (mis)understanding	Experience, self & other, dialogue, genuineness, supportiveness, openness	Source, receiver, signal, information, noise, feedback, redundancy, network, function	Behavior, variable, effect, personality, emotion, perception, cognition, attitude, interaction	Society, structure, practice, ritual, rule, socialization, culture, identity, coconstruction	Ideology, dialectic, oppression, consciousness-raising, resistance, emancipation
Plausible when appeals to metadiscursive commonplaces such as:	Power of words; value of informed judgment; improvability of practice	Understanding requires common language; omnipresent danger of miscommunication	All need human contact, should treat others as persons, respect differences, seek common ground	Identity of mind and brain; value of information and logic; complex systems can be unpredictable	Communication reflects personality; beliefs & feelings bias judgments; people in groups affect one another	The individual is a product of society; every society has a distinct culture; social actions have unintended effects	Self-perpetuation of power & wealth; values of freedom, equality & reason; discussion produces awareness, insight
Interesting when challenges metadiscursive commonplaces such as:	Mere words are not actions; appearance is not reality; style is not substance; opinion is not truth	Words have correct meanings & stand for thoughts; codes & media are neutral channels	Communication is skill; the word is not the thing; facts are objective and values subjective	Humans and machines differ; emotion is not logical; linear order of cause & effect	Humans are rational beings; we know our own minds; we know what we see	Individual agency & responsibility; absolute identity of self; naturalness of the social order	Naturalness & rationality of traditional social order; objectivity of science & technology

Appendix III: Aspect definitions in tool

What/who are the sender(s) and reciever(s) in the contributing factor?			Are the actors in the same space (**local**) or do they communicate remotely (**remote**)?		Does the receiver receive and notice the message right away (**synchronous**) or is there a delay between message produced and message received (**asynchronous**)?	
If both actors are human, then human-human. If at least one actor is human and the other is a machine/computer/device/light or a document/sign, then human-technical. If both actors are machine/computer/device, then machine-machine.			If human-human in same room, then local, if not, then remote. If human-technical in same room, then local, if not (runway lights), then remote. If machine-machine physically connected, then local, if not (ILS-receiver), then remote.		In synchronous communication, the receivers in the communication system process the message from the sender at practically the same time at which the message was produced, there is no or hardly any delay. In asynchronous communication, the receiver does not process the message from the sender at the time it is produced.	
Human-Human (HH)	**Human-Technical (HT)**	**Machine-Machine (MM)**	**Local**	**Remote**	**Synchronous**	**Asynchronous**
E.g. crew-dispatcher, engineer-crew, purser-manager	E.g. ATC-screen, crew - runway lights, engineer-gauge, crew-autopilot, check list-crew, tasksheet-engineer	E.g. yoke-flight controls, MCP-Autopilot, ILS-autopilot	E.g. face-to-face, checklists, display, FADEC-engine	E.g. radio, email, runway lights, ILS-Localizer receiver	E.g. phone, face to face, radio, chat	E.g. checklist, manual, email

Appendices

Do the actors both send messages actively (**interactive**), does only one actor send and the other actor receives (**push**) or does one actor retrieve information on own initiative (**pull, not for human-human**)?			Is the information transmission in the contributing factor via symbols; signs, words, numbers or code (**verbal**) or is it via lights, vocal tone, physical force, bodily movements etc (**nonverbal**)?		Through which sense(s) is the information perceived by the receiver of the information? (Leave the cells null in case of machine-machine interaction)				
Interactive communication is two, or more, actors both functioning as sender and receiver actively. Push communication is a one way method of sending information, where the receiver does not give a notable response (pilot changing altitude, response: right number changes on display). Pull communication is a one way method for retrieving information, such as using a database, book, checklist (does not apply for human-human interaction)			Verbal communication is using words or numbers to convey a message, in either auditory, written or coded form (e.g. morse or coded language). Nonverbal communication is using everything except words or numbers to convey a message, such as force, body language, intonation, touch, symbols, vibration, light, visuals.		This column is only for contributing factors where a human was involved. Multiple senses can be specified. If both actors are non-human, leave the cells null.				
Interactive	**Push**	**Pull (not for HH)**	**Verbal**	**Non-Verbal**	**Sight**	**Hearing**	**Taste**	**Smell**	**Touch**
In human-human interaction, this can be a face to face contact, a phone call, but also a email conversation, as this continues on one subject (this would be asynchronous). In human-technical interaction, interactive could be setting up the FMS or hand flying the aircraft. In machine-machine interaction, this could the constant communication between the ILS and the aircraft.	In human-human interaction, push communication is when an actor only sends information, the receiver not needing to respond. In human-techincal interaction, push communication is when an actor sends information, the receiving technical not needing to provide a response (number changes on display). The same applies to machine-machine interaction (Aileron actuator)	In human-human interaction the pull communication does not apply. In human-technical, pull is the retreival of information from a machine or document source (display, FMS, manual). In machine-machine interaction, pull is the retrieval of information from one machine to another (	E.g. talking on radio, reading manual, seeing aircraft attitude position	E.g. landing lights, mechanical actuator, pushing a yoke.					

What is the communicational media between the two actors in the contributing factor? (not applicable to **local human-human** interaction)							
In this column the communication media of the interaction is specified, which is the channel that was used to convey the message from actor A to B. For a pilot-ATC interaction this will generally be "Radio", for pilot-autopilot							
Radio	**Phone**	**Internet**	**HT Display / Device**	**Infrastructure**	**Document**	**HT aural warning**	**Cable or actuator**
Radio contact between actors (ATC-crew, NAV-beacon)	Phone contact between actors (ATC-airport)	Internet contact between two actors (Skype, ACARS, CPDLC)	A display as principal communication medium, in **human-technical interacton** (Yoke-Pilot, ATC screen-ATC controller,PFD-pilot, NAV-pilot).	Infrastructure as medium of communication (sign, taxi lines, runway lighting)	A document or sign which is read by human (Manual, Taxi sign, checklist)	An human-technical aural message which warns / alerts people of danger (TCAS, EGPWS, Decompression alarm)	A cable or a actuator which is used to convey information amongst machines and computers (ARINC, mechanical actuator)

Appendix IV: Example of contributing factor analysis

		What/who are the sender(s) and reciever(s) in the contributing factor?			Are the actors in the same space (**local**) or do they communicate remotely (**remote**)?		Does the receiver receive and notice the message right away (**synchronous**) or is there a delay between message produced and message received (**asynchronous**)?	
Report	Factors	Human-Human (HH)	Human-Technical (HT)	Machine-Machine (MM)	Local	Remote	Synchronous	Asynchronous
2007015	Factor 1	1				1	1	
Notes->	The crew of the Airbus A319 replied to the permission to take off intended for the Airbus A330.	ATC-A319: The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances				The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances	The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances	

Do the actors both send messages actively (**interactive**), does only one actor send and the other actor receives (**push**) or does one actor retrieve information on own initiative (**pull is applicable to human-human**)?			Is the information transmission in the contributing factor via language; words, numbers or code (**verbal**) or is it via lights, visuals, vocal tone, physical force, bodily movements etc (**nonverbal**)?		Through which sense(s) is the information perceived by the receiver of the information? (Leave the cells null in case of machine-machine interaction)				
Interactive	Push	Pull (not for HH)	Verbal	Non-Verbal	Sight	Hearing	Taste	Smell	Touch
1			1		1				
The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances			The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances			The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances			

What is the communicational media between the two actors in the contributing factor? (not applicable to **local human-human** interaction)								
Radio	Phone	Internet	HT Display /Device	Infrastructure	Document	HT aural warning	Cable or actuator	Remarks
1								
The possibility of confusion in the call signs and the expectation of receiving permission to take off were unfavourable circumstances								

Appendix V: Example of Chi Square test outcome

TIMING * VERB			VERB		Total
			NOVERB	VERB	
TIMING	ASYNC	Count	2	23	25
		% within TIMING	8,00%	92,00%	100,00%
		% within VERB	10,00%	28,00%	24,50%
		% of Total	2,00%	22,50%	24,50%
	SYNC	Count	18	59	77
		% within TIMING	23,40%	76,60%	100,00%
		% within VERB	90,00%	72,00%	75,50%
		% of Total	17,60%	57,80%	75,50%
Total		Count	20	82	102
		% within TIMING	19,60%	80,40%	100,00%
		% within VERB	100,00%	100,00%	100,00%
		% of Total	19,60%	80,40%	100,00%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig.(2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	2,831[a]	1	0,092		
Continuity Correction[b]	1,939	1	0,164		
Likelihood Ratio	3,282	1	0,07		
Fisher's Exact Test				0,146	0,076
N of Valid Cases	102				

a. 1 cells (25,0%) have expected count less than 5. The minimum expected count is 4,90.

b. Computed only for a 2x2 table

1. Aspect one, in this case the form of information, divided in verbal and nonverbal
2. Aspect two, in this case the timing of the information transfer, divided in synchronous and asynchronous
3. Frequency of asynchronous-nonverbal specifications found, in this case 2
4. Percentage of asynchronous nonverbal specifications, opposed to asynchronous verbal specifications, in this case 8.0%
5. Percentage of asynchronous nonverbal specifications, opposed to synchronous nonverbal specifications, in this case 10.0%
6. P value of the Pearson Chi-Square test, in this case 0.092
7. P value of the Fisher's Exact test, in this case 0.146
8. Expected counts, the values of the expected counts determine which test is used

Appendix VI: Syntaxes for SPSS calculations

Actor

```
"CROSSTABS
  /TABLES=ACTORS BY DISTANCE SYNC INTER VERB SIGHT HEAR TOUCH COMM
  /FORMAT=AVALUE TABLES
  /STATISTICS=CHISQ
  /CELLS=COUNT ROW COLUMN TOTAL
  /COUNT ROUND CELL."
```

Distance

```
"CROSSTABS
  /TABLES=DISTANCE BY SYNC INTER VERB SIGHT HEAR TOUCH COMM
  /FORMAT=AVALUE TABLES
  /STATISTICS=CHISQ
  /CELLS=COUNT ROW COLUMN TOTAL
  /COUNT ROUND CELL."
```

Timing (SYNC)

```
"CROSSTABS
  /TABLES=SYNC BY INTER VERB SIGHT HEAR TOUCH COMM
  /FORMAT=AVALUE TABLES
  /STATISTICS=CHISQ
  /CELLS=COUNT ROW COLUMN TOTAL
  /COUNT ROUND CELL."
```

Flow (INTER)

```
"CROSSTABS
  /TABLES=INTER BY VERB SIGHT HEAR TOUCH COMM
  /FORMAT=AVALUE TABLES
  /STATISTICS=CHISQ
  /CELLS=COUNT ROW COLUMN TOTAL
  /COUNT ROUND CELL."
```

Form (VERB)

```
"CROSSTABS
  /TABLES=VERB BY SIGHT HEAR TOUCH COMM
  /FORMAT=AVALUE TABLES
  /STATISTICS=CHISQ
  /CELLS=COUNT ROW COLUMN TOTAL
  /COUNT ROUND CELL."
```

Sight

```
"CROSSTABS
  /TABLES=SIGHT BY HEAR TOUCH COMM
  /FORMAT=AVALUE TABLES
  /STATISTICS=CHISQ
  /CELLS=COUNT ROW COLUMN TOTAL
  /COUNT ROUND CELL."
```

Hearing

```
"CROSSTABS
  /TABLES=HEAR BY TOUCH COMM
  /FORMAT=AVALUE TABLES
  /STATISTICS=CHISQ
  /CELLS=COUNT ROW COLUMN TOTAL
  /COUNT ROUND CELL."
```

Touch

```
"CROSSTABS
  /TABLES=TOUCH BY COMM
  /FORMAT=AVALUE TABLES
  /STATISTICS=CHISQ
  /CELLS=COUNT ROW COLUMN TOTAL
  /COUNT ROUND CELL."
```

Appendix VII: Inter-rater reliability attempt one

Intraclass correlation average

Report	Intraclass correlation
1194718	0,64
1221688	0,95
1245605	0,35
1265538	0,40
1278705	0,86
Average	**0,641**

Results of SPSS calculations

Reliability Statistics[a]

Cronbach's Alpha	N of Items
,879	4

a. REPORT = 1194718

Intraclass Correlation Coefficient[a]

	Intraclass Correlation[c]	95% Confidence Interval		F Test with True Value 0			
		Lower Bound	Upper Bound	Value	df1	df2	Sig
Single Measures	.644[b]	,264	,928	8,242	5	15	,001
Average Measures	,879	,589	,981	8,242	5	15	,001

Reliability Statistics[a]

Cronbach's Alpha	N of Items
,988	4

a. REPORT = 1221688

Intraclass Correlation Coefficient[a]

	Intraclass Correlation[c]	95% Confidence Interval		F Test with True Value 0			
		Lower Bound	Upper Bound	Value	df1	df2	Sig
Single Measures	.948[b]	,838	,992	84,097	5	15	,000
Average Measures	,987	,954	,998	84,097	5	15	,000

Reliability Statistics[a]

Cronbach's Alpha	N of Items
,711	4

a. REPORT = 1245605

Intraclass Correlation Coefficient[a]

	Intraclass Correlation[c]	95% Confidence Interval		F Test with True Value 0			
		Lower Bound	Upper Bound	Value	df1	df2	Sig

Single Measures	.351[b]	,005	,822	3,466	5	15	,028
Average Measures	,684	,020	,949	3,466	5	15	,028

Reliability Statistics[a]

Cronbach's Alpha	N of Items
,711	4

a. REPORT = 1265538

Intraclass Correlation Coefficient[a]

	Intraclass Correlation[c]	95% Confidence Interval		F Test with True Value 0			
		Lower Bound	Upper Bound	Value	df1	df2	Sig
Single Measures	.398[b]	,002	,850	3,462	5	15	,028
Average Measures	,725	,007	,958	3,462	5	15	,028

Reliability Statistics[a]

Cronbach's Alpha	N of Items
,964	4

a. REPORT = 1278705

Intraclass Correlation Coefficient[a]

	Intraclass Correlation[c]	95% Confidence Interval		F Test with True Value 0			
		Lower Bound	Upper Bound	Value	df1	df2	Sig
Single Measures	.864[b]	,630	,977	28,143	5	15	,000
Average Measures	,962	,872	,994	28,143	5	15	,000

Results of analysis by raters

	Actors			Distance			Verbal			Timing				Senses						Communicational media							
Report	1	2	3	1	2	3	1	2	3	1	2	3	4	1	2	3	4	5	6	1	2	3	4	5	6	7	8
Rater 1																											
1265538		1				1	1					1		1													1
1278705		1				1	1					1		1									1				
1194718	1				1		1					1			1					1							
1245605		1		1					1	1				1									1				
1221688	1			1			1					1			1												1
Rater 2																											
1265538	1				1		1			1									1	1							
1278705		1		1			1					1		1									1				
1194718	1				1		1			1					1					1							
1245605		1		1				1		1				1									1				
1221688	1			1			1			1					1												1
Rater 3																											
1265538		1				1	1			1									1								1

1278705	1	1	1		1	1	1
1194718	1	1	1	1	1	1	
1245605	1	1	1	1	1	1	
1221688	1	1	1	1	1		1
Rater 4							
1265538	1	1	1	1	1		1
1278705	1	1	1	1	1	1	
1194718	1	1	1	1	1	1	
1245605	1	1	1	1	1	1	
1221688	1	1	1	1	1		1

Appendix VIII: Inter-rater reliability attempt two

Results of analysis by raters

Report	Intraclass correlation [%]
2004052	0,989
2009073	1
2011057	0,958
Average	**0,982**

Results of SPSS calculations

Reliability Statistics[a]

Cronbach's Alpha	N of Items
,996	3

a. REPORT = 2004052

Intraclass Correlation Coefficient[a]

	Intraclass Correlation[c]	95% Confidence Interval		F Test with True Value 0			
		Lower Bound	Upper Bound	Value	df1	df2	Sig
Single Measures	.989[b]	,974	,996	274,462	13	26	,000
Average Measures	,996	,991	,999	274,462	13	26	,000

Reliability Statistics[a]

Cronbach's Alpha	N of Items
1,000	3

a. REPORT = 2009073

Intraclass Correlation Coefficient[a]

	Intraclass Correlation[c]	95% Confidence Interval		F Test with True Value 0			
		Lower Bound	Upper Bound	Value	df1	df2	Sig
Single Measures	1.000[b]				6		
Average Measures	1,000				6		

Reliability Statistics[a]

Cronbach's Alpha	N of Items
,983	3

a. REPORT = 2011057

Intraclass Correlation Coefficient[a]

	Intraclass Correlation[c]	95% Confidence Interval		F Test with True Value 0			
		Lower Bound	Upper Bound	Value	df1	df2	Sig
Single Measures	.958[b]	,854	,992	59,714	6	12	,000

Average Measures	,986	,946	,997	59,714	6	12	,000

Intraclass correlation average

	Actors			Distance		timing		Flow			Verbal		senses					communicational media							
Report	1	2	3	1	2	1	2	1	2	3	1	2	1	2	3	4	5	1	2	3	4	5	6	7	8
Rater 1																									
2011057		1		1			1	1				1					1				1				
2009073	1				1	1		1			1			1				1							
2004052		1		1		1				1	1		1								1				
2004052		1		1			1			1	1		1										1		
Rater 2																									
2011057		1		1			1	1				1					1				1				
2009073	1				1	1		1			1			1				1							
2004052		1		1		1				1		1	1								1				
2004052		1		1			1			1	1		1										1		
Rater 3																									
2011057		1		1		1			1			1					1				1				
2009073	1				1	1		1			1			1				1							
2004052		1		1		1				1	1		1								1				
2004052		1		1			1			1	1		1										1		

Appendix IX: Frequencies of 'actors' by 'distance' by 'timing'

Combination	Frequency [N]	Frequency [%]
HH, Remote, Synchronous	28	27,2
HT, Local, Synchronous	28	27,2
HT, Local, Asynchronous	14	13,59
HH, Local, Synchronous	12	11,65
HT, Remote, Asynchronous	11	10,68
HT, Remote, Synchronous	9	8,74
MM, Local, Synchronous	1	0,97
HH, Local, Asynchronous	0	0
HH, Remote, Asynchronous	0	0
MM, Local, Asynchronous	0	0
MM, Remote, Asynchronous	0	0
MM, Remote, Asynchronous	0	0

Appendix X: Frequencies 'actors' by 'distance' by 'timing' by 'information flow'

Combination	Frequency [N]	Frequency [%]
HH, Remote, Synchronous, Interactive	26	25,24
HT, Local, Synchronous, Pull	14	13,59
HT, Local, Asynchronous, Pull	12	11,65
HT, Remote, Asynchronous, Pull	10	9,71
HH, Local, Synchronous, Interactive	9	8,74
HT, Local, Synchronous, Interactive	7	6,80
HT, Local, Synchronous, Push	7	6,80
HT, Remote, Synchronous, Pull	7	6,80
HT, Remote, Synchronous, Interactive	2	1,94
HH, Remote, Synchronous, Push	2	1,94
HH, Local, Synchronous, Pull	2	1,94
HT, Local, Asynchronous, Interactive	1	0,97
HH, Local, Synchronous, Push	1	0,97
HT, Local, Asynchronous, Push	1	0,97
HT, Remote, Asynchronous, Push	1	0,97
MM, Local, Synchronous, Push	1	0,97

Appendix XI: Unique specifications found in analysis

#	Specifications	Frequency [N]	Frequency [%]
1	HH, Remote, Synchronous, Interactive, Verbal, Hear, Radio	22	21,36
2	HT, Local, Asynchronous, Pull, Verbal, Sight, Document	10	9,71
3	HH, Local, Synchronous, Interactive, Verbal, Hear	7	6,80
4	HT, Local, Synchronous, Pull, Verbal, Sight, HT Device/Display	7	6,80
5	HT, Remote, Synchronous, Pull, Nonverbal, Sight	5	4,85
6	HT, Local, Synchronous, Interactive, Nonverbal, Sight, Touch, HT Device/Display	4	3,88
7	HT, Remote, Asynchronous, Pull, Verbal, Hear, Radio	3	2,91
8	HT, Remote, Asynchronous, Pull, Verbal, Sight, Document	3	2,91
9	HT, Local, Synchronous, Push, Nonverbal, Touch, HT Device/Display	3	2,91
10	HT, Local, Synchronous, Pull, Verbal, Sight, Document	2	1,94
11	HT, Local, Synchronous, Pull, Verbal, Hearing, Radio	2	1,94
12	HH, Remote, Synchronous, Interactive, Verbal, Hearing	2	1,94
13	HH, Local, Synchronous, Interactive, Verbal, Sight, Hearing	2	1,94
14	HT, Remote, Asynchronous, Pull, Verbal, Sight, Infrastructure	2	1,94
15	HH, Local, Synchronous, Pull, Verbal, Hearing	2	1,94
16	HT, Remote, Synchronous, Interactive, Verbal, Hearing, Radio	2	1,94
17	HH, Local, Synchronous, Push, Verbal, Hearing	1	0,97
18	HT, Local, Synchronous, Push, Verbal, Sight, Touch , HT Device/Display	1	0,97
19	HH, Remote, Synchronous, Interactive, Nonverbal, Hearing, Radio	1	0,97
20	HH, Remote, Synchronous, Push, Verbal, Hearing, Radio	1	0,97
21	HH, Remote, Synchronous, Push, Verbal	1	0,97
22	HT, Remote, Asynchronous, Pull, Nonverbal, Sight	1	0,97
23	HT, Remote, Asynchronous, Pull, Verbal, Sight, Hearing	1	0,97
24	HT, Local, Synchronous, Push, Nonverbal,Sight, Touch, HT Device/Display	1	0,97
25	HT, Remote, Synchronous, Pull, Verbal, Sight	1	0,97
26	HT, Local, Synchronous, Push, Verbal, Sight, HT Device/Display	1	0,97
27	HT, Local, Asynchronous, Pull, Verbal, Sight	1	0,97
28	HT, Local, Synchronous, Interactive, Verbal, Sight, HT Device/Display	1	0,97
29	MM, Local, Synchronous, Push, Verbal, Cable / Actuator	1	0,97
30	HT, Local, Synchronous, Pull, Verbal, Hearing, HT Aural Warning	1	0,97
31	HT, Local, Asynchronous, Pull, Verbal, Sight, HT Device/Display	1	0,97
32	HT, Local, Synchronous, Pull, Nonverbal, Sight, HT Device/Display	1	0,97
33	HT, Local, Synchronous, Interactive, Nonverbal, Touch, HT Device/Display	1	0,97
34	HT, Remote, Asynchronous, Push, Verbal, Sight	1	0,97
35	HT, Local, Asynchronous, Interactive, Nonverbal, Touch, HT Device/Display	1	0,97
36	HH, Remote, Synchronous, Interactive, Nonverbal, Sight	1	0,97
37	HT, Remote, Synchronous, Pull, Verbal, Hearing, Radio	1	0,97
38	HT, Local, Synchronous, Interactive, Verbal, Hearing, HT Device/Display	1	0,97
39	HT, Local, Synchronous, Pull, Verbal, Hearing, HT Device/Display	1	0,97
40	HT, Local, Synchronous, Push, Nonverbal, Sight, HT Device/Display	1	0,97
41	HT, Local, Asynchronous, Push, Verbal, Sight, Document	1	0,97

Appendix XII: All contributing factors found with their respective specification

#	# DSB	Contributing factor summary	Specifications
1	2003071	The English language as spoken by the cabin crew was rated by a number of passengers as poor and difficult to understand.	HH, Local, Synchronous, Interactive, Verbal, Hearing
2	2003071	Although the captain regarded the warning as false he should not have resumed the take-off. Besides that the first officer should have prevented him from doing so in accordance with his statement after the accident. At this stage the crew acted unsafe.	HH, Local, Synchronous, Interactive, Verbal, Hearing
3	2003071	It is very likely that in this situation she overreacted on the controls and applied a large power reduction, thereby creating the onset for a high rate of descent	HH, Local, Synchronous, Interactive, Verbal, Hearing
4	2011061	The combination of low rotor RPM, impact inertia, abrupt lowering of the collective pitch and neutral-to-rear cyclic pitch, caused the blades to cut off the tail boom.	HH, Local, Synchronous, Interactive, Verbal, Hearing
5	2012-105	Despite some uncertainty expressed, the crew did not ask for confirmation from ATC.	HH, Local, Synchronous, Interactive, Verbal, Hearing
6	TK1951	The total time between the activation of the stick shaker and the moment the throttle was set to the maximum thrust position was nine seconds. Simulator tests have shown that the situation could have been recovered and the flight continued if the crew immediately after stick shaker activation had moved the thrust levers to maximum thrust as part of the approach to stall recovery procedure.	HH, Local, Synchronous, Interactive, Verbal, Hearing
7	2005078	The driver of Kievit 2 stopped the vehicle on the runway near the end of runway 24 to clear the remains of an animal from the runway. The controller stated that Kievit 2 did not report this on the frequency, something that is normally done.	HH, Local, Synchronous, Interactive, Verbal, Hearing
8	2006082	"He [captain] reacted by increasing the aircraft's pitch attitude. According to him this should have caused the auto thrust system to add power." AND "It is not clear if and how the first officer monitored the aircraft's instruments during this flight phase. He did not intervene or draw the captain's attention to the speed decay."	HH, Local, Synchronous, Interactive, Verbal, Sight, Hearing
9	2006082	"It is possible that the remark "A/C no CAT III capability" confused the flight crew and let them to believe that the system was serviceable for all other types of flights and approaches. The use of the English language could also have been a factor." AND "The company's technical staff has used the MEL information outside its context and wrote in the maintenance logbook, without explicitly referring to the auto thrust failure: "Managed speed dropping in VLS4 zone on final approach. A/C no CAT III capability"."	HH, Local, Synchronous, Interactive, Verbal, Sight, Hearing
10	98-85/S-14	The available multi-mode screens were in use by the two controllers responsible for the two active runways. The pictures were adjusted for their convenience. The Assistant Controller was therefore not able to check and monitor position and movement of Charlie 8 and the misinterpretation went undiscovered.	HH, Local, Synchronous, Pull, Verbal, Hearing
11	98-85/S-14	He [runway controller] turned to the Assistant Controller for confirmation., heard she was talking to Charlie 8 and assumed that Charlie 8 was reporting clear of runway. He did not wait for further confirmation and cleared DAL39 for take-off.	HH, Local, Synchronous, Pull, Verbal, Hearing
12	98-85/S-14	During the evacuation passengers were not sufficiently instructed on how to leave the aircraft. This prolonged the evacuation.	HH, Local, Synchronous, Push, Verbal, Hearing
13	TK1951	"When the aircraft was descending above Flevoland and just after that, three landing gear configuration warnings were heard. These warnings are not usual during this phase of the flight." AND "The landing gear configuration warnings … could have been a reason for the crew to diagnose the problem. No indication thereof was found on the cockpit voice recorder."	HH, Remote, Synchronous, Interactive, Nonverbal, Hearing, Radio
14	2005142	The controller misunderstood that Kievit 1 wanted to use taxiway Sierra south, not cross runway 06/24. "At 11.25:13 hours Kievit 1 requested clearance to cross runway 06/24 after a departing aircraft on runway 24, which just had passed in front of him. He gave his position as taxiway Sierra south. The tower assistant 24 asked who was calling for Sierra south. When the bird controller identified himself, the assistant 2 replied with the clearance 'Kievit 1, Sierra south approved'."	HH, Remote, Synchronous, Interactive, Nonverbal, Sight
15	98-85/S-14	When the purser initiated the evacuation, the order was not heard in the aft cabin due to the failed P.A. system. This created some confusion and also caused some further delay in the evacuation of the passengers in the aft cabin .	HH, Remote, Synchronous, Interactive, Verbal, Hearing
16	TK1951	In this case the PF could have asked the air traffic controller for permission to use taxiway VS. The controller, in turn, could have provided information on the slipperiness of that taxiway in that case.	HH, Remote, Synchronous, Interactive, Verbal, Hearing
17	98-85/S-14	The bird controllers' instructions do not indicate how often active runways at Amsterdam Airport Schiphol should be inspected. No driving speed provisions have been incorporated either. The established work instructions for bird controllers are insufficiently clear on these points.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio

18	98-85/S-14	"The term "slippery spots" for all taxiways and aprons notably gave the wrong picture of the state of slipperiness of those parts of the taxiway system that had not (yet) been treated with anti-slipperiness means." AND "Both members of the crew indicated that they interpreted the description of the slipperiness as local, small slippery spots."	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
19	98-85/S-14	Though air traffic control performed its task and was entitled to assume its instruction would be followed up, the Dutch Safety Board holds the view that the slipperiness on VS and P6 could have been reported more explicitly to the crew of EZY5112.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
20	2007112	Dutch Safety Board recognizes that ATC personnel should have a certain amount of freedom in handling traffic because of changing situations and the dynamics of the traffic flow. At the same time, this flexibility in handling traffic should be conducted in a safe and orderly manner, within written boundaries.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
21	2007112	The ATC controller guiding the Airbus from the east, did not give the crew a final clearance for the approach to runway 18C. When given a final clearance, the crew is allowed to autonomously turn to the final heading and intercept finals. Because a clearance was not given, the crew persisted on the last given heading and crossed centreline. This caused them to fly west of the centreline of runway 18C.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
22	2006133	The controller missed **four** clues that N666MX was on a "Woody1B" departure; it was mentioned in the first intercom contact with Tower;	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
23	2006133	"It was presented on the Electronic Data Display and it was presented in the label on the monitor" AND "Air traffic control at Rotterdam Airport used full colour monitors with self designed lay out and history dots, which indicate the track flown by aircraft. Air traffic control at Schiphol used standard, round, monochrome monitors without history dots."	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
24	2006133	Not following this instruction played a roll in this serious incident; the approach controller stated the he instructed PH-XRV to descend to 3000 ft to establish sufficient separation between both aircraft which had an opposite course. If PH-XRV had followed the instruction, sufficient vertical separation would have been established.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
25	2012-105	The first clearance from Eindhoven Ground Control where EI-DLD was instructed to leave the runway at intersection Foxtrot, Eindhoven Ground Control did not use the term 'vacate'. Instead the phrase 'intersection Foxtrot to the right' was used.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
26	2012040	The initial taxi instruction given to the crew by the Ground Controller West reads: "taxi via Z to E17". In the instruction the word "via" is. Later on, Ground Controller North instructs the crew to "cross at Z2. Taxi Alpha North to gate E17". This instruction did not state the words "via Z" which, in combination with the position of the aircraft when the clearance was given, namely at the end of runway 18C, may have contributed to the confusion on the flight deck.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
27	2004006_088	After being pushed-back from gate D3 the ground controller cleared the aircraft to 'taxi runway 36C, to leave via A10 and taxi via the south'. When the intersection with taxiway Alpha was reached, the flight crew turned to the left instead of to the right.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
28	2004006_088	This was noticed by the ground controller, who subsequently instructed the flight crew 'at A12 first right now for taxiway Bravo, right and then left at A12, straight ahead'. The clearance was read back as 'Copied, right now to A12, then taxi Bravo, Cabo Verde 625'. The flight crew made a right turn along A11 whereupon the ground controller instructed the Boeing 757 to turn left now for Bravo. The flight crew read back the clearance, turned left on taxiway Bravo and then turned right and entered the active landing runway 36R via E4 (red track on figure 1).	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
29	TK1951	A turn-in, whereby interception takes place at between 6.2 and 5 NM, with no instruction to descend to an altitude below 2000 feet is in deviation of the International Civil Aviation Organization guideline specifying that the aircraft must be flying level on its final approach course before the glide slope is intercepted.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
30	98-85/S-14	The misinterpretation started when in the first call from Charlie 8 to the Assistant Controller no information was given about actual position, requested taxi route and destination. The call contained, conform existing procedures, only a request to cross runway 24 at the second exit. The Assistant Controller did not ask for further clarification and assumed for no specific reason that Charlie 8 was waiting at exit 2W of runway 24 with destination S-apron. This information was passed to the Controller responsible for runway 24.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
31	98-85/S-14	The stopbar control panels were however still configured for the previous set-up of the control tower and his action had no effect. When the following actions to extinguish the stopbar for which Charlie 8 was waiting, failed, the Controller, with the consent of the Supervisor/Coach, pushed the buttons to extinguish the stopbars on both sides i.e. 2W and 2E. The possibility of a wrong interpretation of the position of Charlie 8 was not considered.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
32	2009073	A misinterpretation of a clearance from the ground controller, in the course of which no standard terminology had been applied, resulted in the crew of G-LEAA taxiing onto runway 22 without permission.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio

33	2006088	He [tug driver] gave the signal that all equipment was clear of the aircraft while the tug was still within the aircraft's manoeuvring area. This lead to confusion of the flight crew members. On the other hand the flight crew added power before being sure that the tug was clear of the aircraft.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
34	2007015	The crew of the Airbus A319 replied to the permission to take off intended for the Airbus A330.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
35	2007015	The crew of the Airbus A330 on runway 18L did not reply immediately to the permission to take off.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
36	2004176	The crew was already on the point where they were told to hold short, so they thought the controller meant to hold short of the next runway: "To confirm the clearance with the ground controller, we repeated his command and since there was no correction or objection from his side we began taxiing across runway 22 towards the holding position for the next runway on taxiway G5. Runway 22 was clear of traffic at the time."	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
37	2005142	When the bird controller contacted the tower assistant 2 for the first time the quality of the transmission was very poor. When using a hand walkie-talkie the quality was much better.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
38	2005078	Reduced ability to easilyobserve the vehicle might have been an unfavourable precondition in the mix of illuminated runway and taxi lights during darkness. This also might have contributed to the controller's assumption that the vehicle had vacated the runway.	HH, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
39	2007015	The captain stated that when he wanted to initiate the evacuation he could not find the P.A. handset. Instead he shouted the order to evacuate with the door to the cabin closed. The evacuation order was not heard and therefore not acknowledged. Apart from hearing some commotion going on in the cabin, the captain did not positively ascertain that his order was heard and that the evacuation had started.	HH, Remote, Synchronous, Push, Verbal
40	2005078	In this respect it should be noted that by not establishing a clear and definite crosswind limit in the Transavia Operations Manual a defence barrier against unsafeoperations was lost.	HH, Remote, Synchronous, Push, Verbal, Hearing, Radio
41	2007015	Neither the runway controller who was being relieved nor his relief heard the permission to take off being read back by the crew of the Airbus A319.	HT, Local, Asynchronous, Interactive, Nonverbal, Touch,HT Device / display
42	2010094	The term 'active runway' is not clearly defined in the Air Traffic Control Operations Manual.	HT, Local, Asynchronous, Pull, Verbal, Sight, Document
43	2007015	No procedure has been set out in the Air Traffic Control Operations Manual for the use of the 'runway occupied' strip and for reversing the flight progress strip.	HT, Local, Asynchronous, Pull, Verbal, Sight, Document
44	2007015	Although the bird controller is required to monitor the ATC tower frequency in accordance with his work instructions, the bird controller did not hear the runway controller issue take-off clearance to the departing Boeing 737. This can be explained by the fact that he had turned down the volume slightly on the radio communication equipment to concentrate on inspecting the runway.	HT, Local, Asynchronous, Pull, Verbal, Sight, Document
45	2007015	Shortly thereafter the ATIS message was amended with the wind 230/33, varying between 200 and 260 degrees, and gusts up to 45 kt. This ATIS message was inadvertently transmitted under the same code as the previous one. Because of the same letter code, ATC as well as the cockpit crew were not alerted that the wind conditions had changed.	HT, Local, Asynchronous, Pull, Verbal, Sight, Document
46	2003004	Seat allocation was done by Aviapartner Passenger Services in such a way that passengers were assigned seats in the rear of the cabin to keep the seats in the front of the cabin empty to facilitate the boarding at the next station, Maastricht. This is contradictory to the instruction in the ASM, which prescribes that the seats shall be issued in such a way that an equally spread passenger distribution is obtained.	HT, Local, Asynchronous, Pull, Verbal, Sight, Document
47	2003004	The recorded flight data showed that at a radio height of approximately 50 feet the aircraft was pitched up slightly. This pitch-up could be considered an early flare manoeuvre, which normally occurs around 20 feet above the runway touchdown zone.	HT, Local, Asynchronous, Pull, Verbal, Sight, Document
48	2003004	She again felt unhappy with the situation, which she judged again as too high and too fast. The Captain (PNF) however convinced her to continue the approach at which point she lowered the collective pitch lever and at the same time raised the nose of the helicopter.	HT, Local, Asynchronous, Pull, Verbal, Sight, Document
49	2007112	"The rules and instructions for air traffic services are lacking a clear distinction between strict regulations and general guidelines; for that reason they are ambiguous." AND "The non-compliance with the rules and regulations for VDV and the consequent breaching of essential safety protections were a causal factor in the occurrence."	HT, Local, Asynchronous, Pull, Verbal, Sight, Document
50	2012-105	Later, when EI-DLD was in the vicinity of intersection 'Foxtrot', Eindhoven Ground Control states "this one to the right, and for departure contact tower 131.0, good flight, bye bye".	HT, Local, Asynchronous, Pull, Verbal, Sight, Document
51	2012040	He [captain] recently transitioned to the Airbus A330. Therefore taxiing the aircraft still demanded some additional attention.	HT, Local, Asynchronous, Pull, Verbal, Sight, Document
52	TK1951	As a result of intercepting the glide slope signal from above, the incorrect operation of the autothrottle was obscured for the crew.	HT, Local, Asynchronous, Pull, Verbal, Sight,HT Device / display

53	2007112	In this case no information was shared. The ATC controller for 18C expected his colleague to give him a warning when westerly traffic was approaching again. The ATC controller for 18R was convinced that his colleague would notice new traffic appearing on his radar screen and therefore did not notify him of the Boeing approaching from the west.	HT, Local, Asynchronous, Pull, Verbal, Sight
54	2003014	Most likely the wind notation 120/23 was incorrectly copied from the other (discarded) take-off data card. In addition the maximum wind gust of 28 knots was also not noted.	HT, Local, Asynchronous, Push, Verbal, Sight, Document
55	2007015	It was noisy in the tower and the tower frequencies linked to the runways under the runway controller's responsibility were not being listen to through headphones, but through the tabletop loudspeakers.	HT, Local, Synchronous, Interactive, Nonverbal, Touch,HT Device / display
56	2006092	"Furthermore the wrong wind information reported by L7-Q may have resulted in some confusion with a negative influence on the flight performance." AND "The wrong wind direction of 180º given to the crew by the radio operator of L7-Q was caused by a stuck anemometer."	HT, Local, Synchronous, Interactive, Nonverbal, Sight, Touch,HT Device / display
57	2002124	The investigation and reconstruction of the accident show that shortly after shutting down the left engine, the still functioning right engine also stopped, probably as a result of an unintentional, incorrect manoeuvre by the crew, most likely the closure of the right engine fuel valve.	HT, Local, Synchronous, Interactive, Nonverbal, Sight, Touch,HT Device / display
58	2012040	At the moment the Ground Controller North instructed the crew to "cross at Z2, taxi Alpha North to gate E17', the Airbus A330 was approaching intersection W11 on taxiway Zulu. FO-1 checked his Jeppesen chart of Schiphol in order to locate the route to Alpha North. FO-2 looked over the shoulder of FO-1 to assist him. Both first officers discussed the possible location of Alpha North and were, as a result, not looking outside.	HT, Local, Synchronous, Interactive, Nonverbal, Sight, Touch,HT Device / display
59	2006062	"At 20:32 (13 minutes before landing) the crew announced to air traffic control to have received information Kilo. Just after having received this information, the PNF stated to the PF that he copied a wind from 220° with 15 kt." AND ATIS KILO: "ATIS K 20:30 220° 27kt max35kt, min15kt"	HT, Local, Synchronous, Interactive, Nonverbal, Sight, Touch,HT Device / display
60	2008031	The inadequate look-out of the marshaller prevented an early warning for taxiing traffic behind the aircraft to the pilots of CSDFP.	HT, Local, Synchronous, Interactive, Verbal, Hearing,HT Device / display
61	2006133	the crew of N666MX mentioned the departure when they initially called Approach	HT, Local, Synchronous, Interactive, Verbal, Sight,HT Device / display
62	2007015	The crossing traffic was not transferred to the runway controller. Possible causes for this were the difference in (radio) workload between the runway and the ground controller, a service-oriented attitude towards the pilots, a difference in understanding and application of procedures amongst individual controllers and aspects relating to team resource management.	HT, Local, Synchronous, Pull, Nonverbal, Sight,HT Device / display
63	TK1951	"The crew could have seen on the primary flight display that this remark was not in accordance with the mode status of the autopilot. The primary flight display indicated 'SINGLE CH' (single channel - single autopilot engaged for the ILS approach)." AND "The ... problems when activating the autopilots could have been a reason for the crew to diagnose the problem. No indication thereof was found on the cockpit voice recorder."	HT, Local, Synchronous, Pull, Verbal, Hearing, HT aural warning
64	2003057	"Aircraft N18HJ was equipped with a weather radar which was not switched on during the incident flight. When the aircraft entered the clouds it was too late to switch the weather radar on because, according to the flight crew, it would take 3 till 4 minutes for the system to become active." AND "Although the flight crew responded with the appropriate corrective procedures when they flew through the cloud, the decision not to activate the weather radar can be criticized."	HT, Local, Synchronous, Pull, Verbal, Hearing,HT Device / display
65	2007015	At Amsterdam Airport Schiphol bird controllers are expected to monitor several aviation frequencies simultaneously when carrying out runway inspections. This is an unreasonable requirement.	HT, Local, Synchronous, Pull, Verbal, Hearing, Radio
66	2007015	Moreover, some bird controllers feel they work in a subordinate role to air traffic controllers, which may negatively affect communications.	HT, Local, Synchronous, Pull, Verbal, Hearing, Radio
67	2003071	The standard assigned passenger weight(s) to be used in the load and trim sheet were not unambiguously laid down in the operating manual.	HT, Local, Synchronous, Pull, Verbal, Sight, Document
68	2003071	The DOI used to calculate the TO-CG was not representing the actual situation.	HT, Local, Synchronous, Pull, Verbal, Sight, Document
69	2003071	The 'runway occupied' instruments in the ATC tower do not provide the runway controller any indication of the number of vehicles or aircraft that are (still) occupying the runway.	HT, Local, Synchronous, Pull, Verbal, Sight,HT Device / display
70	2010077	The visual conditions and rain at the time of landing may have impaired the crew's visual depth perception on the runway.	HT, Local, Synchronous, Pull, Verbal, Sight,HT Device / display
71	TK1951	Due to the approach heading and altitude provided to the crew by air traffic control, the localizer signal was intercepted at 5.5 NM from the runway threshold with the result that the glide slope had to be intercepted from above. This obscured the fact that the autothrottle had entered the retard flare mode.	HT, Local, Synchronous, Pull, Verbal, Sight,HT Device / display

72	TK1951	As the airspeed continued to drop, the aircraft's pitch attitude kept increasing. The crew failed to recognise the airspeed decay (INCLUDING WARNINGS) and the pitch increase until the moment the stick shaker was activated.	HT, Local, Synchronous, Pull, Verbal, Sight,HT Device / display
73	TK1951	"The cockpit crew did not have information regarding the interrelationship between the (failure of the) left radio altimeter system and the operation of the autothrottle." AND "With the knowledge available to them at that time, the crew had no way of understanding the actual significance of these indications and warning signals and could not have been expected to determine the pending risk accurately."	HT, Local, Synchronous, Pull, Verbal, Sight,HT Device / display
74	98-85/S-14	Normally the Assistant Controller checks the position of vehicle(s) visually or uses the groundradar to establish the position. The visibility from the control tower was practically zero.	HT, Local, Synchronous, Pull, Verbal, Sight,HT Device / display
75	98-85/S-14	The Supervisor/Coach was aware that a vehicle was about to cross runway 24. Because of the discussion about the stopbar control panel however he probably missed the Take off Clearance being transmitted to DAL39. Thus an overall picture of the traffic situation was missing and he did not intervene.	HT, Local, Synchronous, Pull, Verbal, Sight,HT Device / display
76	2002124	Because not enough attention was paid to the primary task of steering the aircraft, speed dropped below stalling speed. This led to a loss of control of the aircraft at an altitude at which recovery was no longer possible.	HT, Local, Synchronous, Push, Nonverbal, Touch,HT Device / display
77	2012-105	Limitations for airport use that are written down in the AIP were part of the pilots' airfield brief for Eindhoven Airport, but were written in small print, and were therefore not very obvious.	HT, Local, Synchronous, Push, Nonverbal, Touch,HT Device / display
78	2003014	Whilst it is possible that a strong wind gust at V1 could have caused the initial displacement, the wind was not responsible for the runway excursion. At speeds around V1 the rudder authority is more than sufficient to counter any offset caused by the existing wind. Instead the runway excursion was caused by full left rudder input. It could not be determined whether this input was made by the F/O2. The reason for the large rudder pedal input at V1 however could not be determined.	HT, Local, Synchronous, Push, Nonverbal, Touch,HT Device / display
79	2006062	The investigation revealed that the handling of the aircraft was not optimal during the landing phase.	HT, Local, Synchronous, Push, Nonverbal, Sight,HT Device / display
80	TK1951	The consequences of an erroneous radio altimeter signal and the unsafe autothrottle design should have led to the conclusion in 2004 there was a safety problem. An reaction from Boeing would have been to issue a warning to all users. For example a warning can be issued using an 'operations manual bulletin' and is usual in cases where aircraft systems function differently than may be expected. This information should then have been included in the Flight Crew Operations Manual.	HT, Local, Synchronous, Push, Nonverbal, Sight, Touch,HT Device / display
81	2012-105	On the other hand the Co-Pilot – other than the decision to make a go-around after the first approach – failed to let the Captain know in a positive way that she felt uncomfortable.	HT, Local, Synchronous, Push, Verbal, Sight,HT Device / display
82	2006062	However, the standard procedure to engage the ATS was not followed. Instead of stabilizing the engines at 1.40 EPR first and then switch the ATS on, the throttles were manually advanced beyond the position required for the take-off thrust.	HT, Local, Synchronous, Push, Verbal, Sight, Touch,HT Device / display
83	TK1951	Crew resource management and crew communications during the approach were not in accordance with the standard operating procedure of Turkish Airlines regarding cockpit communication.	HT, Remote, Asynchronous, Pull, Nonverbal, Sight,Infrastructure
84	97-75/A-26	During the final approach the crew received a final wind reading from the tower of 240/30 kt maximum 43 kt, resulting in a crosswind of 35 kt. The cockpit crew understood the maximum gust value as "forty" in stead of "four three".	HT, Remote, Asynchronous, Pull, Verbal, Hearing, Radio
85	2003133	It is striking that there is a substantial difference in time between the slipperiness situation reported on by the AAS and the time this was included in the ATIS-information. The Aim's reporting indicates that at 18:30 hours there were already slippery spots on the taxiways and aprons, whereas the request to the ATC to include the warning in the CCIS and in the ATIS information was only made at 19:39 hours. Because initially the content of the warning was not included in its entirety in the message of 19:39 hours. "All aprons slippery spots", it was only at 20:14 hours that a warning came about "All aprons and taxiways slippery spots". It took about one hour and 45 minutes to make this information known to the airmen.	HT, Remote, Asynchronous, Pull, Verbal, Hearing, Radio
86	2006062	The flap setting of 40° was contrary to the initial intention to land with flaps 28°. It has not become clear why the PF changed his decision to land with flaps 40 in stead of flaps 28. He did not explain this to the PNF nor did the PNF ask for the reason for this change. Also the determination of the wind components was not discussed thoroughly and did probably not lead to a full understanding of the conditions that could be expected during final approach and landing.	HT, Remote, Asynchronous, Pull, Verbal, Hearing, Radio

87	TK1951	The assumption that air traffic control "would not come there anyway" turned out to be incorrect, this appeared to be the area beyond intersection V3. The AOM did not communicate this decision to air traffic control. Therefore, in principle, ATC was not aware of the impracticability of VS, P6 and P7. This is not in compliance with the Dutch civil aviation regulations (RTL), which states that the airport manager must promptly inform ATC of which part of the landing area can be used.	HT, Remote, Asynchronous, Pull, Verbal, Sight, Document
88	TK1951	The procedure regarding passenger distribution in the Handbook Cabin Personnel was different from the procedure in the CSM. This could lead to uncertainty by the responsible cabin crew member concerning which procedure was applicable.	HT, Remote, Asynchronous, Pull, Verbal, Sight, Document
89	2012-105	The crew were aware of work in progress at the airfield, they were not aware of the content of the memo sent by Eindhoven Airport Authorities to all operators with the details about interim taxi procedures.	HT, Remote, Asynchronous, Pull, Verbal, Sight, Document
90	2012040	On the relevant Jeppesen chart the area around intersection W11 and W12 is indicated as “Hot spot”.9 Separately, highlighted on the chart, a warning is incorporated not to turn left near W11 or W12 when taxiing on taxi track Zulu coming from the North via holding point Z2. The more detailed Jeppesen chart (10-9A) on which the hotspot area is enlarged was missed by the crew and therefore not used.	HT, Remote, Asynchronous, Pull, Verbal, Sight,Infrastructure
91	2004006_088	The crew of the Boeing 757 did not stop for the hold line. Visibility in relation to precipitation and/or combined with light reflection cannot be excluded as unfavourable preconditions to clearly see the hold line. Runway 36R was used for landing.	HT, Remote, Asynchronous, Pull, Verbal, Sight,Infrastructure
92	2012040	The procedure in the BOM referring to the distribution of passengers did not give criteria on how to check if passengers were evenly distributed and how to correct deviations. The indication that reseating could be necessary is considered to be insufficient with the absence of instructions on how to accomplish this.	HT, Remote, Asynchronous, Pull, Verbal, Sight, Hearing
93	2011057	The investigation concludes that the control problems were probably the result of a mis-trimmed aircraft.	HT, Remote, Asynchronous, Push, Verbal, Sight
94	2005142	The bird controller also indicated that some of the multiple radio channels he had to monitor sometimes blocked out each other or were of such a poor quality that this may have had a negative effect on his monitoring performance.	HT, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
95	2003014	ATIS ‘S’ was not received by the crew, neither was the change in the ATIS reports mentioned by ATC, however, when ATC issued the take-off clearance, wind information was provided which indicated a crosswind in excess of 30 knots (320/18 maximum 34).	HT, Remote, Synchronous, Interactive, Verbal, Hearing, Radio
96	2007044	"Prior to descent the crew recorded EHGG ATIS message ‘G’, stating runway 23 in use and a surface wind of 200 degrees at 10 knots, variable between 170 and 230 degrees." AND "The radar plot shows a recorded ground speed of 266 knots at 5 NM from the threshold. From the flight data recorder an indicated airspeed of 222 knots was derived, equating to 226 knots true air speed at 2000 feet. This implies that the aircraft was under the influence of a 40 knots tail wind component (266 knots groundspeed minus 226 knots true airspeed) and shows that the tail wind was of greater influence on the descent than the crew expected or realised. During the interviews neither pilot mentioned that the approach was affected by tail wind."	HT, Remote, Synchronous, Pull, Nonverbal, Sight
97	2005142	The relationship within the crew was more instructor/student than PF/PNF related. It is considered likely that the Captain by frequently taking the initiative created a situation whereby the Co-Pilot had to work faster than her experience would justify.	HT, Remote, Synchronous, Pull, Nonverbal, Sight
98	2007112	The radar images shown to the ATC controllers has a 5 seconds lag time. This lag is due to system design and is not preventable. Because of radar lag time and the short time between the given instructions the ATC controller was unaware of the fact the crew was already turning to the instructed headings.	HT, Remote, Synchronous, Pull, Nonverbal, Sight
99	2004006_088	The ground control monitoring process did not prevent that the Boeing 757 entered and taxied on runway 36R. Assumptions, other tasks and traffic, procedures and runways in use were identified as (possible) unfavourable preconditions.	HT, Remote, Synchronous, Pull, Nonverbal, Sight
100	2005078	The visual check by the flight crew did not work. The surrounding lights might be an unfavourable precondition to observe a vehicle at 3500 meter distance during darkness.	HT, Remote, Synchronous, Pull, Nonverbal, Sight
101	2008031	Insufficient provision of information to the pilots of CS-DFP regarding to the procedure for a full power engine run-up test, resulting in non-adherence to the proper procedure .	HT, Remote, Synchronous, Pull, Verbal, Hearing, Radio
102	2011051	The most probable cause of the door separation was the door handle not being in the locked position in combination with excessive free play in the forward door hinge.	HT, Remote, Synchronous, Pull, Verbal, Sight
103	TK1951	The operating software designed to compare the two radio altimeter systems cannot be applied in the entire Boeing 737 NG fleet. The introduction of operating software capable of making comparisons has failed to fully eliminate the undesired activation of the retard flare mode.	MM, Local, Synchronous, Push, Verbal,Cable

Appendix XIII: Elaborations of the 36 aspect vs. aspect combinations

Actors - Distance

This combination has a significant relationship (P=0.000). Remote was found significantly more frequent in human - human interactions (70.0%) and local was found significantly more frequent in human –technical interactions (67.7%). Local was found significantly less frequent in human - human interactions (30.0%) and remote was found significantly less frequent in human –technical interactions (32.2%).

			DISTANCE		
			LOC	REM	Total
ACTORS	HH	Count	12	28	40
		% within ACTORS	30,0%	70,0%	100,0%
		% within DISTANCE	22,2%	58,3%	39,2%
		% of Total	11,8%	27,5%	39,2%
	HT	Count	42	20	62
		% within ACTORS	67,7%	32,3%	100,0%
		% within DISTANCE	77,8%	41,7%	60,8%
		% of Total	41,2%	19,6%	60,8%
Total		Count	54	48	102
		% within ACTORS	52,9%	47,1%	100,0%
		% within DISTANCE	100,0%	100,0%	100,0%
		% of Total	52,9%	47,1%	100,0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	13,902[a]	1	0,000		
Continuity Correction[b]	12,428	1	0,000		
Likelihood Ratio	14,209	1	0,000		
Fisher's Exact Test				0,000	0,000
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 18,82.

b. Computed only for a 2x2 table

Actors - Timing

This combination has a significant relationship (P=0.000). Synchronous was found significantly more frequent in human - human interactions (100.0%) and synchronous was also found significantly more frequent in human –technical interactions (59.7%). Asynchronous was found significantly less frequent in human - human interactions (0.0%) and asynchronous was also found significantly less frequent in human –technical interactions (40.3%).

			SYNC		
			ASYNC	SYNC	Total
ACTORS	HH	Count	0	40	40
		% within ACTORS	0.0%	100.0%	100.0%
		% within SYNC	0.0%	51.9%	39.2%
		% of Total	0.0%	39.2%	39.2%
	HT	Count	25	37	62
		% within ACTORS	40.3%	59.7%	100.0%
		% within SYNC	100.0%	48.1%	60.8%
		% of Total	24.5%	36.3%	60.8%
Total		Count	25	77	102
		% within ACTORS	24.5%	75.5%	100.0%
		% within SYNC	100.0%	100.0%	100.0%
		% of Total	24.5%	75.5%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	21,366[a]	1	0.000		
Continuity Correction[b]	19.242	1	0.000		
Likelihood Ratio	29.992	1	0.000		
Fisher's Exact Test				0.000	0.000
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 9,80.

b. Computed only for a 2x2 table

Actors - Information flow

This combination has a significant relationship (P=0.000). Interactive was found significantly more frequent in human - human interactions (87.5%) and pull was found significantly more frequent in human –technical interactions (69.4%). Pull (5.0%) and push (7.5%) were found significantly less frequent in human - human interactions and interactive (16.1%) and push (14.5%) were also found significantly less frequent in human –technical interaction.

			INTER			
			INT	PULL	PUSH	Total
ACTORS	HH	Count	35	2	3	40
		% within ACTORS	87.5%	5.0%	7.5%	100.0%
		% within INTER	77.8%	4.4%	25.0%	39.2%
		% of Total	34.3%	2.0%	2.9%	39.2%
	HT	Count	10	43	9	62
		% within ACTORS	16.1%	69.4%	14.5%	100.0%
		% within INTER	22.2%	95.6%	75.0%	60.8%
		% of Total	9.8%	42.2%	8.8%	60.8%
Total		Count	45	45	12	102
		% within ACTORS	44.1%	44.1%	11.8%	100.0%
		% within INTER	100.0%	100.0%	100.0%	100.0%
		% of Total	44.1%	44.1%	11.8%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	51,914[a]	2	0.000
Likelihood Ratio	59.086	2	0.000
N of Valid Cases	102		

a. 1 cells (16,7%) have expected count less than 5. The minimum expected count is 4,71.

Actors - Verbal

This combination has a significant relationship (P=0.003). Verbal was found significantly more frequent in human - human interactions (95.0%) and verbal was also found significantly more frequent in human –technical interactions (71.0%). Nonverbal was found significantly less frequent in human - human interactions (5.0%) and was also found significantly less frequent in human –technical interactions (29.0%).

			VERB		
			NOVERB	VERB	Total
ACTORS	HH	Count	2	38	40
		% within ACTORS	5.0%	95.0%	100.0%
		% within VERB	10.0%	46.3%	39.2%
		% of Total	2.0%	37.3%	39.2%
	HT	Count	18	44	62
		% within ACTORS	29.0%	71.0%	100.0%
		% within VERB	90.0%	53.7%	60.8%
		% of Total	17.6%	43.1%	60.8%
Total		Count	20	82	102
		% within ACTORS	19.6%	80.4%	100.0%
		% within VERB	100.0%	100.0%	100.0%
		% of Total	19.6%	80.4%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	8,908[a]	1	0.003		
Continuity Correction[b]	7.449	1	0.006		
Likelihood Ratio	10.379	1	0.001		
Fisher's Exact Test				0.004	0.002
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 7,84.

b. Computed only for a 2x2 table

Actors - Sight

This combination has a significant relationship (P=0.000). Sight was found significantly less frequent in human - human interactions (92.5%) and sight was found significantly more frequent in human –technical interactions (74.2%).

			SIGHT		
			NO	YES	Total
ACTORS	HH	Count	37	3	40
		% within ACTORS	92.5%	7.5%	100.0%
		% within SIGHT	69.8%	6.1%	39.2%
		% of Total	36.3%	2.9%	39.2%
	HT	Count	16	46	62
		% within ACTORS	25.8%	74.2%	100.0%
		% within SIGHT	30.2%	93.9%	60.8%
		% of Total	15.7%	45.1%	60.8%
Total		Count	53	49	102
		% within ACTORS	52.0%	48.0%	100.0%
		% within SIGHT	100.0%	100.0%	100.0%
		% of Total	52.0%	48.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	43,326[a]	1	0.000		
Continuity Correction[b]	40.695	1	0.000		
Likelihood Ratio	49.128	1	0.000		
Fisher's Exact Test				0.000	0.000
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 19,22.

b. Computed only for a 2x2 table

Actors - Hearing

This combination has a significant relationship (P=0.000). Hearing was found significantly more frequent in human - human interactions (95.0%) and hearing was also found significantly less frequent in human –technical interactions (80.6%).

			HEAR		
			NO	YES	Total
ACTORS	HH	Count	2	38	40
		% within ACTORS	5.0%	95.0%	100.0%
		% within HEAR	3.8%	76.0%	39.2%
		% of Total	2.0%	37.3%	39.2%
	HT	Count	50	12	62
		% within ACTORS	80.6%	19.4%	100.0%
		% within HEAR	96.2%	24.0%	60.8%
		% of Total	49.0%	11.8%	60.8%
Total		Count	52	50	102
		% within ACTORS	51.0%	49.0%	100.0%
		% within HEAR	100.0%	100.0%	100.0%
		% of Total	51.0%	49.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	55,673[a]	1	0.000		
Continuity Correction[b]	52.687	1	0.000		
Likelihood Ratio	64.557	1	0.000		
Fisher's Exact Test				0.000	0.000
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 19,61.

b. Computed only for a 2x2 table

Actors - Touch

This combination has a significant relationship (P=0.005). Touch was found significantly less frequent in human - human interactions (100.0%) and was found significantly more frequent in human –technical interactions (17.7%).

			TOUCH		
			NO	YES	Total
ACTORS	HH	Count	40	0	40
		% within ACTORS	100.0%	0.0%	100.0%
		% within TOUCH	44.0%	0.0%	39.2%
		% of Total	39.2%	0.0%	39.2%
	HT	Count	51	11	62
		% within ACTORS	82.3%	17.7%	100.0%
		% within TOUCH	56.0%	100.0%	60.8%
		% of Total	50.0%	10.8%	60.8%
Total		Count	91	11	102
		% within ACTORS	89.2%	10.8%	100.0%
		% within TOUCH	100.0%	100.0%	100.0%
		% of Total	89.2%	10.8%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	7,955[a]	1	0.005		
Continuity Correction[b]	6.217	1	0.013		
Likelihood Ratio	11.800	1	0.001		
Fisher's Exact Test				0.003	0.003
N of Valid Cases	102				

a. 1 cells (25,0%) have expected count less than 5. The minimum expected count is 4,31.

b. Computed only for a 2x2 table

Actors - Communicational media

This combination has a significant relationship (P=0.000). Radio (60.0%) and none specified (40.0%) were found significantly more frequent in human - human interactions and document (25.8%) and HT device (40.3%) were found significantly more frequent in human –technical interaction. Document and HT device were not present in human - human interactions. None specified (21.0%) and radio (12.9%) was present in human - technical interactions.

			COMM				
				DOCU	HTDEV	RAD	Total
ACTORS	HH	Count	16	0	0	24	40
		% within ACTORS	40.0%	0.0%	0.0%	60.0%	100.0%
		% within COMM	55.2%	0.0%	0.0%	75.0%	39.2%
		% of Total	15.7%	0.0%	0.0%	23.5%	39.2%
	HT	Count	13	16	25	8	62
		% within ACTORS	21.0%	25.8%	40.3%	12.9%	100.0%
		% within COMM	44.8%	100.0%	100.0%	25.0%	60.8%
		% of Total	12.7%	15.7%	24.5%	7.8%	60.8%
Total		Count	29	16	25	32	102
		% within ACTORS	28.4%	15.7%	24.5%	31.4%	100.0%
		% within COMM	100.0%	100.0%	100.0%	100.0%	100.0%
		% of Total	28.4%	15.7%	24.5%	31.4%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	46,740[a]	3	0.000
Likelihood Ratio	60.738	3	0.000
N of Valid Cases	102		

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 6,27.

Distance - Timing

This combination does not have a significant relationship (P= 0.724). Interactive was found frequent in both local (74.1%) as remote interactions (77.1%).

			SYNC		
			ASYNC	SYNC	Total
DISTANCE	LOC	Count	14	40	54
		% within DISTANCE	25.9%	74.1%	100.0%
		% within SYNC	56.0%	51.9%	52.9%
		% of Total	13.7%	39.2%	52.9%
	REM	Count	11	37	48
		% within DISTANCE	22.9%	77.1%	100.0%
		% within SYNC	44.0%	48.1%	47.1%
		% of Total	10.8%	36.3%	47.1%
Total		Count	25	77	102
		% within DISTANCE	24.5%	75.5%	100.0%
		% within SYNC	100.0%	100.0%	100.0%
		% of Total	24.5%	75.5%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	,124[a]	1	0.724		
Continuity Correction[b]	0.015	1	0.903		
Likelihood Ratio	0.125	1	0.724		
Fisher's Exact Test				0.819	0.452
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 11,76.

b. Computed only for a 2x2 table

Distance - Information flow

This combination has a significant relationship (P=0.018). Pull was found significantly more frequent in local interactions (51.9%) and interactive was also found significantly more frequent in remote interactions (58.3%). Push (16.7%) and interactive (31.5%) were found significantly less frequent in local interactions and push (35.4%) and pull (6.3%) were found significantly less frequent in remote interactions.

			INTER			
			INT	PULL	PUSH	Total
DISTANCE	LOC	Count	17	28	9	54
		% within DISTANCE	31.5%	51.9%	16.7%	100.0%
		% within INTER	37.8%	62.2%	75.0%	52.9%
		% of Total	16.7%	27.5%	8.8%	52.9%
	REM	Count	28	17	3	48
		% within DISTANCE	58.3%	35.4%	6.3%	100.0%
		% within INTER	62.2%	37.8%	25.0%	47.1%
		% of Total	27.5%	16.7%	2.9%	47.1%
Total		Count	45	45	12	102
		% within DISTANCE	44.1%	44.1%	11.8%	100.0%
		% within INTER	100.0%	100.0%	100.0%	100.0%
		% of Total	44.1%	44.1%	11.8%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	8,053[a]	2	0.018
Likelihood Ratio	8.219	2	0.016
N of Valid Cases	102		

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 5,65.

Distance - Verbal

This combination does not have significant relationship (P=0.481). Verbal was found frequent in both local interactions (77.8%) as remote interactions (83.3%).

			VERB		
			NOVERB	VERB	Total
DISTANCE	LOC	Count	12	42	54
		% within DISTANCE	22.2%	77.8%	100.0%
		% within VERB	60.0%	51.2%	52.9%
		% of Total	11.8%	41.2%	52.9%
	REM	Count	8	40	48
		% within DISTANCE	16.7%	83.3%	100.0%
		% within VERB	40.0%	48.8%	47.1%
		% of Total	7.8%	39.2%	47.1%
Total		Count	20	82	102
		% within DISTANCE	19.6%	80.4%	100.0%
		% within VERB	100.0%	100.0%	100.0%
		% of Total	19.6%	80.4%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	,498[a]	1	0.481		
Continuity Correction[b]	0.208	1	0.649		
Likelihood Ratio	0.501	1	0.479		
Fisher's Exact Test				0.619	0.326
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 9,41.

b. Computed only for a 2x2 table

Distance - Sight

This combination has a significant relationship (P=0.001). Sight was found significantly more frequent in local interactions (63.0%) and significantly less frequent in remote interactions (31.3%).

			SIGHT		
			NO	YES	Total
DISTANCE	LOC	Count	20	34	54
		% within DISTANCE	37.0%	63.0%	100.0%
		% within SIGHT	37.7%	69.4%	52.9%
		% of Total	19.6%	33.3%	52.9%
	REM	Count	33	15	48
		% within DISTANCE	68.8%	31.3%	100.0%
		% within SIGHT	62.3%	30.6%	47.1%
		% of Total	32.4%	14.7%	47.1%
Total		Count	53	49	102
		% within DISTANCE	52.0%	48.0%	100.0%
		% within SIGHT	100.0%	100.0%	100.0%
		% of Total	52.0%	48.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	10,239[a]	1	0.001		
Continuity Correction[b]	9.007	1	0.003		
Likelihood Ratio	10.432	1	0.001		
Fisher's Exact Test				0.002	0.001
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 23,06.

b. Computed only for a 2x2 table

Distance - Hearing

This combination has a significant relationship (P=0.000). Hearing was found significantly less frequent in local interactions (31.5%) and significantly more frequent in remote interactions (31.3%).

			HEAR		
			NO	YES	Total
DISTANCE	LOC	Count	37	17	54
		% within DISTANCE	68.5%	31.5%	100.0%
		% within HEAR	71.2%	34.0%	52.9%
		% of Total	36.3%	16.7%	52.9%
	REM	Count	15	33	48
		% within DISTANCE	31.3%	68.8%	100.0%
		% within HEAR	28.8%	66.0%	47.1%
		% of Total	14.7%	32.4%	47.1%
Total		Count	52	50	102
		% within DISTANCE	51.0%	49.0%	100.0%
		% within HEAR	100.0%	100.0%	100.0%
		% of Total	51.0%	49.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	14,124[a]	1	0.000		
Continuity Correction[b]	12.672	1	0.000		
Likelihood Ratio	14.465	1	0.000		
Fisher's Exact Test				0.000	0.000
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 23,53.

b. Computed only for a 2x2 table

Distance - Touch

This combination has a significant relationship (P=0.001). Touch was found significantly more frequent in local interactions (20.4%) and significantly less frequent in remote interactions (100.0 %).

			TOUCH		
			NO	YES	Total
DISTANCE	LOC	Count	43	11	54
		% within DISTANCE	79.6%	20.4%	100.0%
		% within TOUCH	47.3%	100.0%	52.9%
		% of Total	42.2%	10.8%	52.9%
	REM	Count	48	0	48
		% within DISTANCE	100.0%	0.0%	100.0%
		% within TOUCH	52.7%	0.0%	47.1%
		% of Total	47.1%	0.0%	47.1%
Total		Count	91	11	102
		% within DISTANCE	89.2%	10.8%	100.0%
		% within TOUCH	100.0%	100.0%	100.0%
		% of Total	89.2%	10.8%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	10,960[a]	1	0.001		
Continuity Correction[b]	8.945	1	0.003		
Likelihood Ratio	15.171	1	0.000		
Fisher's Exact Test				0.001	0.001
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 5,18.

b. Computed only for a 2x2 table

Distance - Communicational media

This combination has a significant relationship (P=0.000). Document (24.1%) and HT device (46.3%) were found significantly more frequent in local interactions and radio (62.5%) and none specified (31.3%) were found significantly more frequent in remote interactions.

			COMM				
				DOCU	HTDEV	RAD	Total
DISTANCE	LOC	Count	14	13	25	2	54
		% within DISTANCE	25.9%	24.1%	46.3%	3.7%	100.0%
		% within COMM	48.3%	81.3%	100.0%	6.3%	52.9%
		% of Total	13.7%	12.7%	24.5%	2.0%	52.9%
	REM	Count	15	3	0	30	48
		% within DISTANCE	31.3%	6.3%	0.0%	62.5%	100.0%
		% within COMM	51.7%	18.8%	0.0%	93.8%	47.1%
		% of Total	14.7%	2.9%	0.0%	29.4%	47.1%
Total		Count	29	16	25	32	102
		% within DISTANCE	28.4%	15.7%	24.5%	31.4%	100.0%
		% within COMM	100.0%	100.0%	100.0%	100.0%	100.0%
		% of Total	28.4%	15.7%	24.5%	31.4%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	55,624[a]	3	0.000
Likelihood Ratio	70.476	3	0.000
N of Valid Cases	102		

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 7,53.

Timing - Information flow

This combination has a significant relationship (P=0.000). interactive was found significantly more frequent in synchronous interactions (57.1%) and pull was found significantly more frequent in asynchronous interactions (88.0%). Push (13.7%) and pull (29.9 %) were found significantly less frequent in synchronous interactions and interactive (4.0%) and push (8.0%) were found significantly less frequent in asynchronous interactions.

			INTER			
			INT	PULL	PUSH	Total
SYNC	ASYNC	Count	1	22	2	25
		% within SYNC	4.0%	88.0%	8.0%	100.0%
		% within INTER	2.2%	48.9%	16.7%	24.5%
		% of Total	1.0%	21.6%	2.0%	24.5%
	SYNC	Count	44	23	10	77
		% within SYNC	57.1%	29.9%	13.0%	100.0%
		% within INTER	97.8%	51.1%	83.3%	75.5%
		% of Total	43.1%	22.5%	9.8%	75.5%
Total		Count	45	45	12	102
		% within SYNC	44.1%	44.1%	11.8%	100.0%
		% within INTER	100.0%	100.0%	100.0%	100.0%
		% of Total	44.1%	44.1%	11.8%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	26,935[a]	2	0.000
Likelihood Ratio	30.839	2	0.000
N of Valid Cases	102		

a. 1 cells (16,7%) have expected count less than 5. The minimum expected count is 2,94.

Timing - Verbal

This combination does not have a significant relationship (P=0.146). Verbal was found frequent in both synchronous interactions (76.6%) as asynchronous interactions (92.0%).

			VERB		
			NOVERB	VERB	Total
SYNC	ASYNC	Count	2	23	25
		% within SYNC	8.0%	92.0%	100.0%
		% within VERB	10.0%	28.0%	24.5%
		% of Total	2.0%	22.5%	24.5%
	SYNC	Count	18	59	77
		% within SYNC	23.4%	76.6%	100.0%
		% within VERB	90.0%	72.0%	75.5%
		% of Total	17.6%	57.8%	75.5%
Total		Count	20	82	102
		% within SYNC	19.6%	80.4%	100.0%
		% within VERB	100.0%	100.0%	100.0%
		% of Total	19.6%	80.4%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	2,831[a]	1	0.092		
Continuity Correction[b]	1.939	1	0.164		
Likelihood Ratio	3.282	1	0.070		
Fisher's Exact Test				0.146	0.076
N of Valid Cases	102				

a. 1 cells (25,0%) have expected count less than 5. The minimum expected count is 4,90.

b. Computed only for a 2x2 table

Timing - Sight

This combination has a significant relationship (P=0.000). Sight was found significantly more frequent in asynchronous interactions (84.0%) and significantly less frequent in synchronous interactions (63.6%).

			SIGHT		
			NO	YES	Total
SYNC	ASYNC	Count	4	21	25
		% within SYNC	16.0%	84.0%	100.0%
		% within SIGHT	7.5%	42.9%	24.5%
		% of Total	3.9%	20.6%	24.5%
	SYNC	Count	49	28	77
		% within SYNC	63.6%	36.4%	100.0%
		% within SIGHT	92.5%	57.1%	75.5%
		% of Total	48.0%	27.5%	75.5%
Total		Count	53	49	102
		% within SYNC	52.0%	48.0%	100.0%
		% within SIGHT	100.0%	100.0%	100.0%
		% of Total	52.0%	48.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	17,157[a]	1	0.000		
Continuity Correction[b]	15.301	1	0.000		
Likelihood Ratio	18.317	1	0.000		
Fisher's Exact Test				0.000	0.000
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 12,01.

b. Computed only for a 2x2 table

Timing - Hearing

This combination has a significant relationship (P=0.000). Hearing was found significantly more frequent in synchronous interactions (59.7%) and significantly less frequent in asynchronous interactions (84.0%).

			HEAR		
			NO	YES	Total
SYNC	ASYNC	Count	21	4	25
		% within SYNC	84.0%	16.0%	100.0%
		% within HEAR	40.4%	8.0%	24.5%
		% of Total	20.6%	3.9%	24.5%
	SYNC	Count	31	46	77
		% within SYNC	40.3%	59.7%	100.0%
		% within HEAR	59.6%	92.0%	75.5%
		% of Total	30.4%	45.1%	75.5%
Total		Count	52	50	102
		% within SYNC	51.0%	49.0%	100.0%
		% within HEAR	100.0%	100.0%	100.0%
		% of Total	51.0%	49.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	14,448[a]	1	0.000		
Continuity Correction[b]	12.751	1	0.000		
Likelihood Ratio	15.575	1	0.000		
Fisher's Exact Test				0.000	0.000
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 12,25.

b. Computed only for a 2x2 table

Timing - Touch

This combination does not have a significant relationship (P=0.286). Touch was not found frequent in both synchronous interactions (87.0%) as asynchronous interactions (96.0%).

			TOUCH		
			NO	YES	Total
SYNC	ASYNC	Count	24	1	25
		% within SYNC	96.0%	4.0%	100.0%
		% within TOUCH	26.4%	9.1%	24.5%
		% of Total	23.5%	1.0%	24.5%
	SYNC	Count	67	10	77
		% within SYNC	87.0%	13.0%	100.0%
		% within TOUCH	73.6%	90.9%	75.5%
		% of Total	65.7%	9.8%	75.5%
Total		Count	91	11	102
		% within SYNC	89.2%	10.8%	100.0%
		% within TOUCH	100.0%	100.0%	100.0%
		% of Total	89.2%	10.8%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	1,584[a]	1	0.208		
Continuity Correction[b]	0.788	1	0.375		
Likelihood Ratio	1.902	1	0.168		
Fisher's Exact Test				0.286	0.191
N of Valid Cases	102				

a. 1 cells (25,0%) have expected count less than 5. The minimum expected count is 2,70.

b. Computed only for a 2x2 table

Timing - Communicational media

This combination has a significant relationship (P=0.000). Radio (37.7%) and HT device (29.9%) were found significantly more frequent in synchronous interactions and document (56.0%) was found significantly more frequent in asynchronous interactions.

			COMM			
			DOCU	HTDEV	RAD	Total
SYNC	ASYNC	Count	6	14	2	3	25
		% within SYNC	24.0%	56.0%	8.0%	12.0%	100.0%
		% within COMM	20.7%	87.5%	8.0%	9.4%	24.5%
		% of Total	5.9%	13.7%	2.0%	2.9%	24.5%
	SYNC	Count	23	2	23	29	77
		% within SYNC	29.9%	2.6%	29.9%	37.7%	100.0%
		% within COMM	79.3%	12.5%	92.0%	90.6%	75.5%
		% of Total	22.5%	2.0%	22.5%	28.4%	75.5%
Total		Count	29	16	25	32	102
		% within SYNC	28.4%	15.7%	24.5%	31.4%	100.0%
		% within COMM	100.0%	100.0%	100.0%	100.0%	100.0%
		% of Total	28.4%	15.7%	24.5%	31.4%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	42,184[a]	3	0.000
Likelihood Ratio	38.128	3	0.000
N of Valid Cases	102		

a. 1 cells (12,5%) have expected count less than 5. The minimum expected count is 3,92.

Information flow - Verbal

This combination does not have a significant relationship (P=0.118). Verbal was found frequent in interactive interactions (82.2%), pull interactions (84.4%) and push interactions (58.3%).

			VERB		
			NOVERB	VERB	Total
INTER	INT	Count	8	37	45
		% within INTER	17.8%	82.2%	100.0%
		% within VERB	40.0%	45.1%	44.1%
		% of Total	7.8%	36.3%	44.1%
	PULL	Count	7	38	45
		% within INTER	15.6%	84.4%	100.0%
		% within VERB	35.0%	46.3%	44.1%
		% of Total	6.9%	37.3%	44.1%
	PUSH	Count	5	7	12
		% within INTER	41.7%	58.3%	100.0%
		% within VERB	25.0%	8.5%	11.8%
		% of Total	4.9%	6.9%	11.8%
Total		Count	20	82	102
		% within INTER	19.6%	80.4%	100.0%
		% within VERB	100.0%	100.0%	100.0%
		% of Total	19.6%	80.4%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	4,269[a]	2	0.118
Likelihood Ratio	3.642	2	0.162
N of Valid Cases	102		

a. 1 cells (16,7%) have expected count less than 5. The minimum expected count is 2,35.

Information flow - Sight

This combination has a significant relationship (P=0.000). Sight was found significantly more frequent in pull interactions (77.8 %) and significantly less frequent in interactive interactions (82.2%). In push interactions, sight was not frequently more found than it was not found (50.0%).

			SIGHT		
			NO	YES	Total
INTER	INT	Count	37	8	45
		% within INTER	82.2%	17.8%	100.0%
		% within SIGHT	69.8%	16.3%	44.1%
		% of Total	36.3%	7.8%	44.1%
	PULL	Count	10	35	45
		% within INTER	22.2%	77.8%	100.0%
		% within SIGHT	18.9%	71.4%	44.1%
		% of Total	9.8%	34.3%	44.1%
	PUSH	Count	6	6	12
		% within INTER	50.0%	50.0%	100.0%
		% within SIGHT	11.3%	12.2%	11.8%
		% of Total	5.9%	5.9%	11.8%
Total		Count	53	49	102
		% within INTER	52.0%	48.0%	100.0%
		% within SIGHT	100.0%	100.0%	100.0%
		% of Total	52.0%	48.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	32,471[a]	2	0.000
Likelihood Ratio	34.815	2	0.000
N of Valid Cases	102		

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 5,76.

Information flow - Hearing

This combination has a significant relationship (P=0.000). Hearing was found significantly more frequent in interactive interactions (82.2%) and significantly less frequent in push (83.3%) and pull (75.6%) interactions.

			HEAR		
			NO	YES	Total
INTER	INT	Count	8	37	45
		% within INTER	17.8%	82.2%	100.0%
		% within HEAR	15.4%	74.0%	44.1%
		% of Total	7.8%	36.3%	44.1%
	PULL	Count	34	11	45
		% within INTER	75.6%	24.4%	100.0%
		% within HEAR	65.4%	22.0%	44.1%
		% of Total	33.3%	10.8%	44.1%
	PUSH	Count	10	2	12
		% within INTER	83.3%	16.7%	100.0%
		% within HEAR	19.2%	4.0%	11.8%
		% of Total	9.8%	2.0%	11.8%
Total		Count	52	50	102
		% within INTER	51.0%	49.0%	100.0%
		% within HEAR	100.0%	100.0%	100.0%
		% of Total	51.0%	49.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	35,752[a]	2	0.000
Likelihood Ratio	38.375	2	0.000
N of Valid Cases	102		

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 5,88.

Information flow - Touch

This combination has insufficient data for a statistical analysis to be performed. Touch was found in eleven contributing factors (10.8%). It was found five times in push interaction (41.7%), 0 times in pull interactions (0.0%) and six times in interactive interactions (13.3%).

			TOUCH		
			NO	YES	Total
INTER	INT	Count	39	6	45
		% within INTER	86.7%	13.3%	100.0%
		% within TOUCH	42.9%	54.5%	44.1%
		% of Total	38.2%	5.9%	44.1%
	PULL	Count	45	0	45
		% within INTER	100.0%	0.0%	100.0%
		% within TOUCH	49.5%	0.0%	44.1%
		% of Total	44.1%	0.0%	44.1%
	PUSH	Count	7	5	12
		% within INTER	58.3%	41.7%	100.0%
		% within TOUCH	7.7%	45.5%	11.8%
		% of Total	6.9%	4.9%	11.8%
Total		Count	91	11	102
		% within INTER	89.2%	10.8%	100.0%
		% within TOUCH	100.0%	100.0%	100.0%
		% of Total	89.2%	10.8%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	17,639[a]	2	0.000
Likelihood Ratio	18.123	2	0.000
N of Valid Cases	102		

a. 3 cells (50,0%) have expected count less than 5. The minimum expected count is 1,29.

Information flow - Communicational media

This combination has insufficient data for a statistical analysis to be performed. In nonverbal interactions, none specified was found 7 times (35%), document 0 times (0.0%), HT device 12 times (60%) and radio was found 1 times (5.0%). In verbal interactions, none specified was found 22 times (26.8%), document 16 times (19.5%), HT device 13 times (15.9%) and radio was found 31 times (37.8%).

			COMM				
				DOCU	HTDEV	RAD	Total
INTER	INT	Count	12	0	8	25	45
		% within INTER	26.7%	0.0%	17.8%	55.6%	100.0%
		% within COMM	41.4%	0.0%	32.0%	78.1%	44.1%
		% of Total	11.8%	0.0%	7.8%	24.5%	44.1%
	PULL	Count	14	15	10	6	45
		% within INTER	31.1%	33.3%	22.2%	13.3%	100.0%
		% within COMM	48.3%	93.8%	40.0%	18.8%	44.1%
		% of Total	13.7%	14.7%	9.8%	5.9%	44.1%
	PUSH	Count	3	1	7	1	12
		% within INTER	25.0%	8.3%	58.3%	8.3%	100.0%
		% within COMM	10.3%	6.3%	28.0%	3.1%	11.8%
		% of Total	2.9%	1.0%	6.9%	1.0%	11.8%
Total		Count	29	16	25	32	102
		% within INTER	28.4%	15.7%	24.5%	31.4%	100.0%
		% within COMM	100.0%	100.0%	100.0%	100.0%	100.0%
		% of Total	28.4%	15.7%	24.5%	31.4%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	38,235[a]	6	0.000
Likelihood Ratio	42.256	6	0.000
N of Valid Cases	102		

a. 4 cells (33,3%) have expected count less than 5. The minimum expected count is 1,88.

Verbal - Sight

This combination has a significant relationship (P=0.028). Sight was found significantly more frequent in nonverbal interactions (70.0%) and significantly less frequent in verbal interactions (42.7%).

			SIGHT		
			NO	YES	Total
VERB	NOVERB	Count	6	14	20
		% within VERB	30.0%	70.0%	100.0%
		% within SIGHT	11.3%	28.6%	19.6%
		% of Total	5.9%	13.7%	19.6%
	VERB	Count	47	35	82
		% within VERB	57.3%	42.7%	100.0%
		% within SIGHT	88.7%	71.4%	80.4%
		% of Total	46.1%	34.3%	80.4%
Total		Count	53	49	102
		% within VERB	52.0%	48.0%	100.0%
		% within SIGHT	100.0%	100.0%	100.0%
		% of Total	52.0%	48.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	4,807[a]	1	0.028		
Continuity Correction[b]	3.775	1	0.052		
Likelihood Ratio	4.897	1	0.027		
Fisher's Exact Test				0.044	0.025
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 9,61.

b. Computed only for a 2x2 table

Verbal - Hearing

This combination has a significant relationship (P=0.000). Hearing was found significantly more frequent in verbal interactions (59.8%) and significantly less frequent in nonverbal interactions (5.0%).

			HEAR		
			NO	YES	Total
VERB	NOVERB	Count	19	1	20
		% within VERB	95.0%	5.0%	100.0%
		% within HEAR	36.5%	2.0%	19.6%
		% of Total	18.6%	1.0%	19.6%
	VERB	Count	33	49	82
		% within VERB	40.2%	59.8%	100.0%
		% within HEAR	63.5%	98.0%	80.4%
		% of Total	32.4%	48.0%	80.4%
Total		Count	52	50	102
		% within VERB	51.0%	49.0%	100.0%
		% within HEAR	100.0%	100.0%	100.0%
		% of Total	51.0%	49.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	19,290[a]	1	0.000		
Continuity Correction[b]	17.161	1	0.000		
Likelihood Ratio	22.888	1	0.000		
Fisher's Exact Test				0.000	0.000
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 9,80.

b. Computed only for a 2x2 table

Verbal - Touch

This combination has a significant relationship (P=0.000). Touch was found significantly more frequent in nonverbal interactions (59.8%) and significantly less frequent in verbal interactions (1.2%).

			TOUCH		
			NO	YES	Total
VERB	NOVERB	Count	10	10	20
		% within VERB	50.0%	50.0%	100.0%
		% within TOUCH	11.0%	90.9%	19.6%
		% of Total	9.8%	9.8%	19.6%
	VERB	Count	81	1	82
		% within VERB	98.8%	1.2%	100.0%
		% within TOUCH	89.0%	9.1%	80.4%
		% of Total	79.4%	1.0%	80.4%
Total		Count	91	11	102
		% within VERB	89.2%	10.8%	100.0%
		% within TOUCH	100.0%	100.0%	100.0%
		% of Total	89.2%	10.8%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	39,765[a]	1	0.000		
Continuity Correction[b]	34.857	1	0.000		
Likelihood Ratio	31.237	1	0.000		
Fisher's Exact Test				0.000	0.000
N of Valid Cases	102				

a. 1 cells (25,0%) have expected count less than 5. The minimum expected count is 2,16.

b. Computed only for a 2x2 table

Verbal - Communicational media

This combination has insufficient data for a statistical analysis to be performed. In document (19.5%), HT device (15.9%), radio (37.8%) and none specified (26.8%) verbal communication was found.

			COMM				Total
				DOCU	HTDEV	RAD	
VERB	NOVERB	Count	7	0	12	1	20
		% within VERB	35,0%	0,0%	60,0%	5,0%	100,0%
		% within COMM	24,1%	0,0%	48,0%	3,1%	19,6%
		% of Total	6,9%	0,0%	11,8%	1,0%	19,6%
	VERB	Count	22	16	13	31	82
		% within VERB	26,8%	19,5%	15,9%	37,8%	100,0%
		% within COMM	75,9%	100,0%	52,0%	96,9%	80,4%
		% of Total	21,6%	15,7%	12,7%	30,4%	80,4%
Total		Count	29	16	25	32	102
		% within VERB	28,4%	15,7%	24,5%	31,4%	100,0%
		% within COMM	100,0%	100,0%	100,0%	100,0%	100,0%
		% of Total	28,4%	15,7%	24,5%	31,4%	100,0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	22,580[a]	3	0,000
Likelihood Ratio	25,391	3	0,000
N of Valid Cases	102		

a. 2 cells (25,0%) have expected count less than 5. The minimum expected count is 3,14.

Sight – Hearing

This combination has a significant relationship (P=0.028). Sight was found significantly more frequent in nonverbal interactions (70.0%) and significantly less frequent in verbal interactions (42.7%).

			HEAR		Total
			NO	YES	
SIGHT	NO	Count	6	47	53
		% within SIGHT	11.3%	88.7%	100.0%
		% within HEAR	11.5%	94.0%	52.0%
		% of Total	5.9%	46.1%	52.0%
	YES	Count	46	3	49
		% within SIGHT	93.9%	6.1%	100.0%
		% within HEAR	88.5%	6.0%	48.0%
		% of Total	45.1%	2.9%	48.0%
Total		Count	52	50	102
		% within SIGHT	51.0%	49.0%	100.0%
		% within HEAR	100.0%	100.0%	100.0%
		% of Total	51.0%	49.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	69,439[a]	1	0.000		
Continuity Correction[b]	66.175	1	0.000		
Likelihood Ratio	81.355	1	0.000		
Fisher's Exact Test				0.000	0.000
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 24,02.

b. Computed only for a 2x2 table

Sight – Touch

This combination does not have a significant relationship (P=0.647). Touch was not found frequent in both sight interactions (12.2%) and no sight interactions (9.4%).

			TOUCH		
			NO	YES	Total
SIGHT	NO	Count	48	5	53
		% within SIGHT	90.6%	9.4%	100.0%
		% within TOUCH	52.7%	45.5%	52.0%
		% of Total	47.1%	4.9%	52.0%
	YES	Count	43	6	49
		% within SIGHT	87.8%	12.2%	100.0%
		% within TOUCH	47.3%	54.5%	48.0%
		% of Total	42.2%	5.9%	48.0%
Total		Count	91	11	102
		% within SIGHT	89.2%	10.8%	100.0%
		% within TOUCH	100.0%	100.0%	100.0%
		% of Total	89.2%	10.8%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	,209[a]	1	0.647		
Continuity Correction[b]	0.019	1	0.890		
Likelihood Ratio	0.209	1	0.648		
Fisher's Exact Test				0.754	0.444
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 5,28.

b. Computed only for a 2x2 table

Sight - Communicational media

This combination has a significant relationship (P=0.000). Document (32.7%), none specified (30.6%) and HT device (36.7%) was found significantly more frequent in sight interactions and radio was significantly less frequent in no sight interactions (60.4%).

			COMM				
			DOCU	HTDEV	RAD	Total
SIGHT	NO	Count	14	0	7	32	53
		% within SIGHT	26.4%	0.0%	13.2%	60.4%	100.0%
		% within COMM	48.3%	0.0%	28.0%	100.0%	52.0%
		% of Total	13.7%	0.0%	6.9%	31.4%	52.0%
	YES	Count	15	16	18	0	49
		% within SIGHT	30.6%	32.7%	36.7%	0.0%	100.0%
		% within COMM	51.7%	100.0%	72.0%	0.0%	48.0%
		% of Total	14.7%	15.7%	17.6%	0.0%	48.0%
Total		Count	29	16	25	32	102
		% within SIGHT	28.4%	15.7%	24.5%	31.4%	100.0%
		% within COMM	100.0%	100.0%	100.0%	100.0%	100.0%
		% of Total	28.4%	15.7%	24.5%	31.4%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	52,799[a]	3	0.000
Likelihood Ratio	71.429	3	0.000
N of Valid Cases	102		

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 7,69.

Hearing – Touch

This combination has a significant relationship (P=0.001). Touch was found significantly less frequent in hearing interactions (100.0%) and significantly more frequent in no hearing interactions (21.2%).

			TOUCH		
			NO	YES	Total
HEAR	NO	Count	41	11	52
		% within HEAR	78.8%	21.2%	100.0%
		% within TOUCH	45.1%	100.0%	51.0%
		% of Total	40.2%	10.8%	51.0%
	YES	Count	50	0	50
		% within HEAR	100.0%	0.0%	100.0%
		% within TOUCH	54.9%	0.0%	49.0%
		% of Total	49.0%	0.0%	49.0%
Total		Count	91	11	102
		% within HEAR	89.2%	10.8%	100.0%
		% within TOUCH	100.0%	100.0%	100.0%
		% of Total	89.2%	10.8%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	11,855[a]	1	0.001		
Continuity Correction[b]	9.759	1	0.002		
Likelihood Ratio	16.102	1	0.000		
Fisher's Exact Test				0.001	0.000
N of Valid Cases	102				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 5,39.

b. Computed only for a 2x2 table

Hearing - Communicational media

This combination has a significant relationship (P=0.001). Radio (64%) and none specified (32.0%) were found significantly more frequent in hearing interactions and document (30.8%) and HT device were significantly less frequent in no hearing interactions.

			COMM				
				DOCU	HTDEV	RAD	Total
HEAR	NO	Count	13	16	23	0	52
		% within HEAR	25.0%	30.8%	44.2%	0.0%	100.0%
		% within COMM	44.8%	100.0%	92.0%	0.0%	51.0%
		% of Total	12.7%	15.7%	22.5%	0.0%	51.0%
	YES	Count	16	0	2	32	50
		% within HEAR	32.0%	0.0%	4.0%	64.0%	100.0%
		% within COMM	55.2%	0.0%	8.0%	100.0%	49.0%
		% of Total	15.7%	0.0%	2.0%	31.4%	49.0%
Total		Count	29	16	25	32	102
		% within HEAR	28.4%	15.7%	24.5%	31.4%	100.0%
		% within COMM	100.0%	100.0%	100.0%	100.0%	100.0%
		% of Total	28.4%	15.7%	24.5%	31.4%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	65,936[a]	3	0.000
Likelihood Ratio	87.533	3	0.000
N of Valid Cases	102		

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 7,84.

Touch - Communicational media

This combination has insufficient data for a statistical analysis to be performed. Touch was found eleven times in HT device interactions (100.0%).

			COMM				Total
				DOCU	HTDEV	RAD	
TOUCH	NO	Count	29	16	14	32	91
		% within TOUCH	31,9%	17,6%	15,4%	35,2%	100,0%
		% within COMM	100,0%	100,0%	56,0%	100,0%	89,2%
		% of Total	28,4%	15,7%	13,7%	31,4%	89,2%
	YES	Count	0	0	11	0	11
		% within TOUCH	0,0%	0,0%	100,0%	0,0%	100,0%
		% within COMM	0,0%	0,0%	44,0%	0,0%	10,8%
		% of Total	0,0%	0,0%	10,8%	0,0%	10,8%
Total		Count	29	16	25	32	102
		% within TOUCH	28,4%	15,7%	24,5%	31,4%	100,0%
		% within COMM	100,0%	100,0%	100,0%	100,0%	100,0%
		% of Total	28,4%	15,7%	24,5%	31,4%	100,0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	37,975[a]	3	0,000
Likelihood Ratio	35,468	3	0,000
N of Valid Cases	102		

a. 4 cells (50,0%) have expected count less than 5. The minimum expected count is 1,73.